

SuperPaymaster: A Decentralized and UX-Optimized Ethereum Gas Payment System Based on Account Abstraction

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Keywords

Blockchain, Ethereum, ERC-4337, Account Abstraction, Paymaster, SeamlessGas Payment, NFT, Transaction Fee, Transaction Security, Standardized Decentralized Service System, SDSS

Highlights

- We provide a comprehensive overview of existing gas payment systems on the Ethereum blockchain and analyze their inherent weaknesses.
- We establish key guidelines and requirements for the design of a decentralized, seamless, and cost-effective gas payment system.
- We propose a novel gas payment (sponsor) system, leveraging the Ethereum platform and an application sponsorship model to address costly and complex processes.
- We conduct a proof of concept (PoC) to demonstrate the advantages of the proposed system over existing solutions and to outline potential future research directions.

Abstract

Current blockchain gas payments impede widespread adoption due to high costs, complexity, and poor user experience (UX)[19,21,22] rooted in Human Computer Interaction(HCI)[7,8,9,10] challenges. While Account Abstraction (ERC-4337)[2] offers potential, centralized implementations with improvements often introduce critical risks like censorship and price manipulation, undermining decentralization. This paper introduces SuperPaymaster, a novel gas payment system using ERC-4337 Paymaster[4] and a Standardized Decentralized Service System (SDSS) to create a truly decentralized, competitive cost-effective, and user-friendly system. It directly tackles high costs[18], usability friction[16], and centralization vulnerabilities[15]. SuperPaymaster provides an open-source framework enabling permissionless Paymaster nodes via a unified contract, fostering competition, supporting diverse ERC-20 gas tokens, and integrating with secure accounts like AirAccount via SDSS for streamlined, secure interactions. By optimizing gas payments through enhanced UX and decentralization, SuperPaymaster aims to significantly lower entry barriers, improve blockchain interaction efficiency

and usability, and ultimately accelerate Web3 adoption[21]. A Proof-of-Concept (PoC) demonstrates the system's feasibility and potential advantages.

1. Introduction

Blockchain technology, underpinning a digital asset market valued in the trillions [23], offers transformative potential for societal collaboration. However, its path to mass adoption is significantly hindered by user experience challenges, particularly surrounding the fundamental process of paying transaction fees, or "gas" on networks like Ethereum. As established by the Technology Acceptance Model (TAM)[24,25]and Human-Computer Interaction (HCI)[7,8,9,10,11,26,27] principles, perceived ease of use is critical for adoption. Currently, interacting with blockchain often involves a cumbersome and costly gas payment process, presenting substantial barriers related to usability, efficiency, and cost[28]. Users, especially newcomers, confront a steep learning curve involving numerous complex steps (often 7 steps, including exchange interactions, KYC, wallet setup, and cross-chain transfers and more), high cognitive load, and opaque procedures which are "Norman's gulf"[36,29]. These usability hurdles act as significant deterrents, preventing many from engaging with and realizing the value of decentralized applications (dApps). The individual wallet address graph show that the user of Ethereum is growing and hit 300 Million in figure 2.

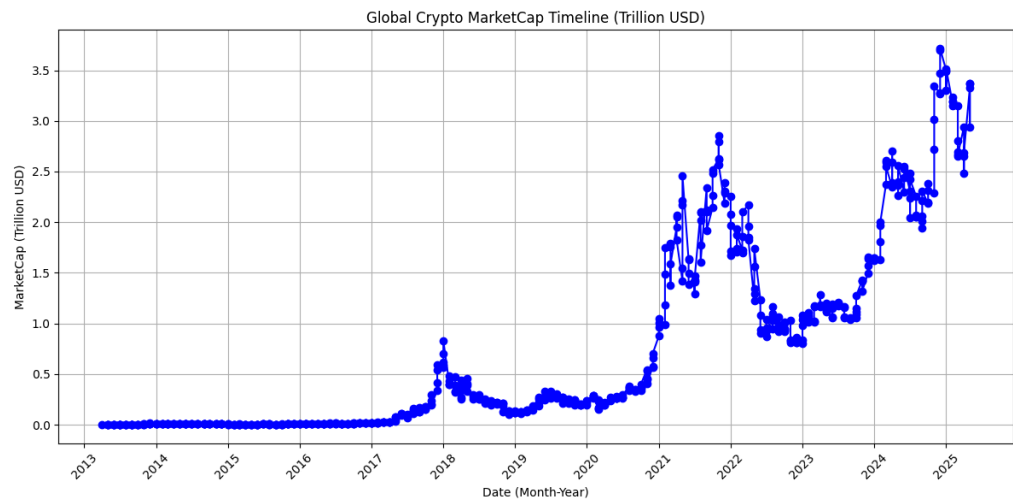


Figure 1: The crypto market cap is valued in the trillions(data source: CoinMarketCap).

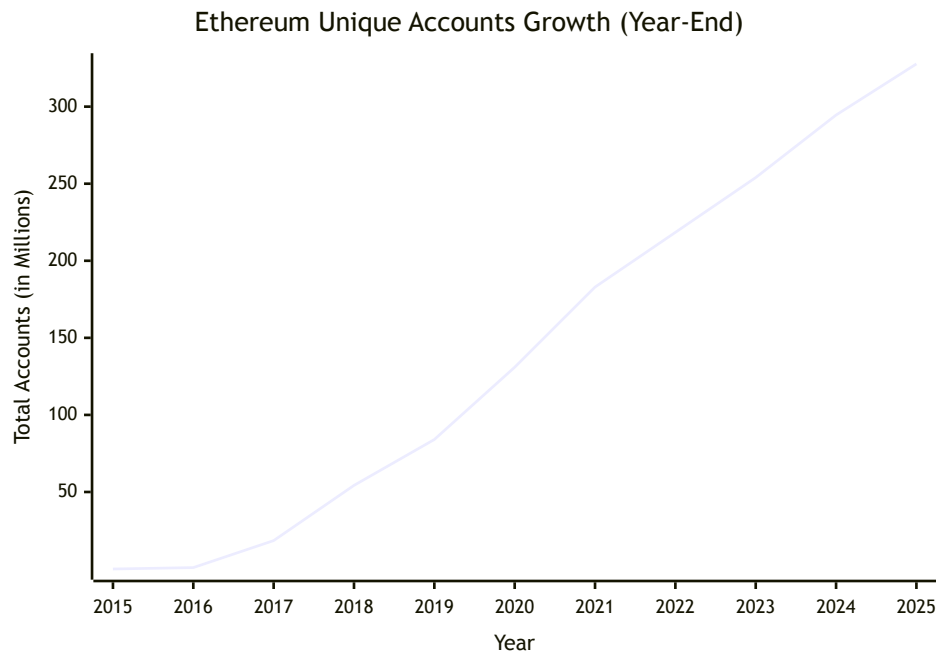


Figure 2: The number of individual wallet addresses on Ethereum is growing and hit 300 Million[777].

The emergence of Account Abstraction (AA), particularly Ethereum's ERC-4337 standard [2], offers promising mechanisms like gas sponsorship (Paymaster) to alleviate some of these burdens. However, current implementations leveraging AA frequently introduce new centralization risks. Many rely on a limited number of centralized entities acting as Bundlers or Paymasters[6]. This approach reintroduces vulnerabilities such as transaction censorship, potential price manipulation by dominant players, and single points of failure, directly conflicting with blockchain's core decentralization capabilities "without a trusted party"[30,31]. Furthermore, practical limitations persist in these centralized solutions, including restricted support for diverse ERC-20 tokens as gas payment, lack of truly permissionless service operation, and complex integration efforts for dApp developers, leaving a critical gap for a genuinely decentralized alternative.

In this paper, we introduce SuperPaymaster, a gas payment system based on ERC-4337 Account Abstraction and a novel Standardized Decentralized Service System (SDSS) architecture. SuperPaymaster is designed to foster a truly decentralized, competitive, and user-friendly system for managing transaction fees. It directly addresses the limitations of previous approaches by enabling an open-source framework where anyone can permissionlessly operate Paymaster nodes. These nodes register via SDSS (using ENS for discovery) and compete to offer gas sponsorship, facilitating lower costs and accepting a wide variety of community-issued or standard ERC-20 tokens. Integration with user-centric wallets like AirAccount further enhances usability and security, aiming for a seamless payment experience. By decentralizing the paymaster layer and prioritizing user experience through intuitive design principles that leverage familiar user paradigms, SuperPaymaster seeks to significantly lower entry barriers, improve interaction efficiency, and accelerate the broader adoption of Web3 technologies. We create a team in two years ago and delivery a Proof-of-Concept (PoC) to test the system's feasibility. Figure 3 from the a16z State of Crypto Report (2024) highlights a critical milestone in the widespread adoption of Web3, reflecting a significant increase in user engagement.

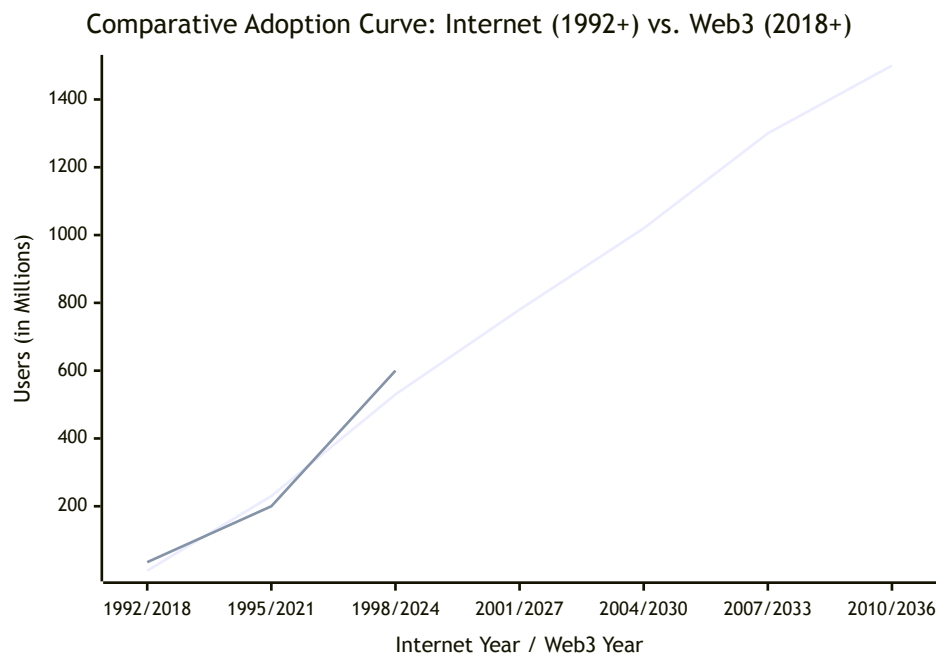


Figure 3: a16z state-of-crypto-report-2024 for web3 rising users [80]

2. Analysis of Current Gas Payment Systems and Associated Challenges

This section delves into the foundational aspects of gas payment mechanisms within EVM-compatible blockchains, outlines the typical user workflow, and critically examines the multifaceted challenges and vulnerabilities inherent in current systems, including usability barriers and the specific risks associated with centralized solutions.

2.1 The Gas Payment Mechanism

2.1.1 Necessity of Gas Payment

The requirement for users to pay 'gas' for transactions is fundamental to the operation and security of public, permissionless blockchains like Ethereum. Due to the Turing-completeness of the Ethereum Virtual Machine (EVM)[13], which allows for arbitrary computation, a mechanism is needed to prevent infinite loops and denial-of-service (DoS) attacks that could exhaust network resources [13]. Gas acts as a computational metering unit, assigning a cost to each operational step executed by the EVM. By requiring payment for computation, the gas mechanism ensures the sustainable use of shared public resources, prevents network abuse, and incentivizes validators (miners/stakers) to process transactions and secure the network[14].

2.1.2 Current Gas Payment Workflow

Executing a transaction on current blockchain systems typically involves a complex, multi-step workflow for the user, even before the core on-chain interaction occurs. A user often needs to: (1) Create an account on a centralized exchange (CEX); (2) Complete Know Your Customer (KYC) verification; (3) Purchase the blockchain's native token (e.g., ETH for Ethereum) using fiat currency; (4) Set up a self-custodial wallet (e.g., MetaMask); (5) Transfer the native token from the CEX to their wallet, incurring withdrawal fees; (6) Potentially perform cross-chain swaps or bridging if operating on a different network or requiring specific tokens, adding further complexity and cost; (7) Finally, initiate the desired transaction (e.g., interacting with a dApp or wallet), requiring careful setting of gas parameters (gas limit, gas price/priority fee) and signing with their private key. This intricate process serves as a significant initial barrier, particularly for non-technical users. As we draw below, there will be over 7 steps in real world comparing with SuperPaymaster with 4 steps(s1,s2 is one time setup, s3 submit transaction, s4 view transaction result).

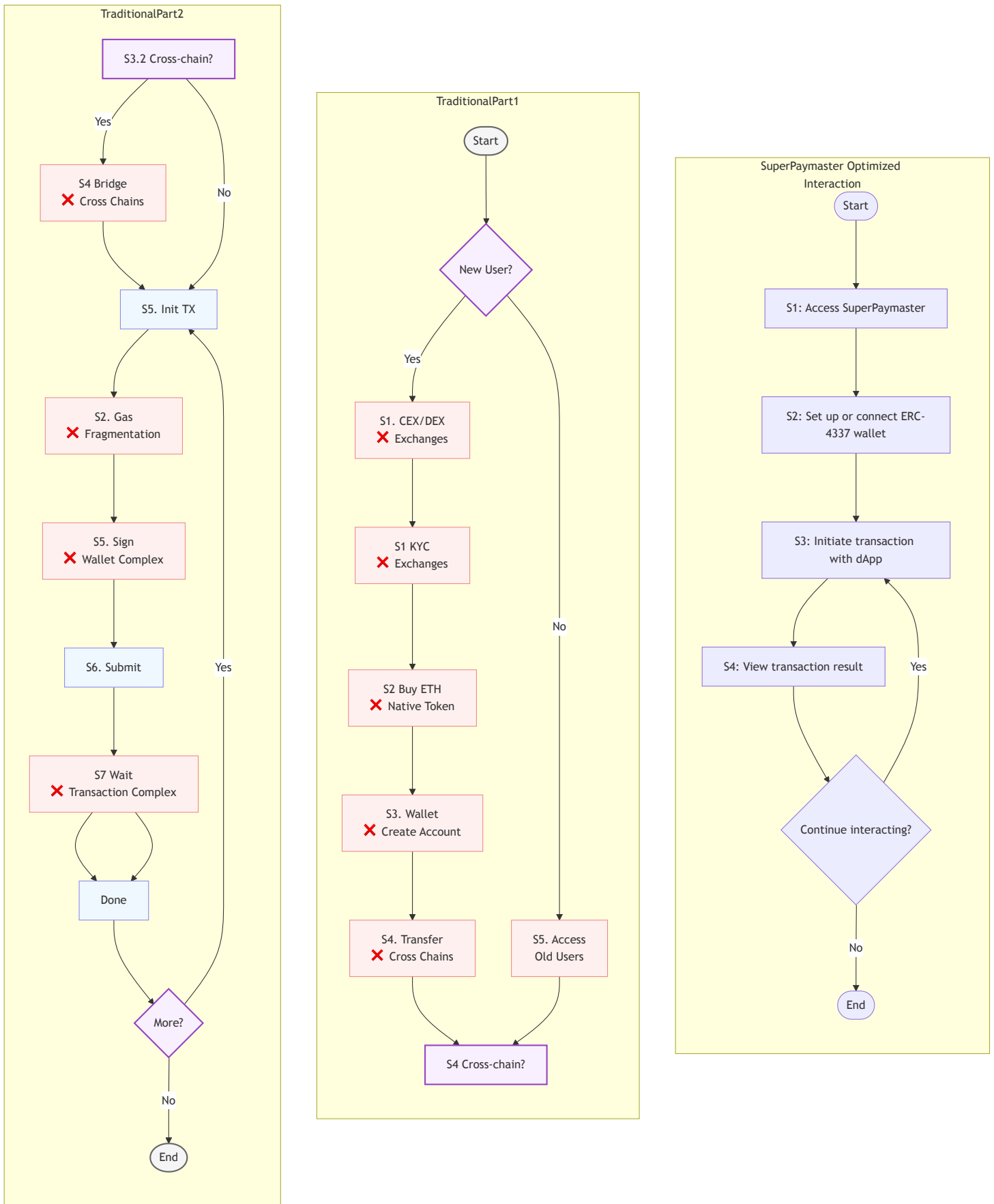


Figure 2: Paymaster and SuperPaymaster System Flow Overview

2.2 Challenges and Vulnerabilities in Current Systems

Existing gas payment systems suffer from a confluence of issues that impede usability, efficiency, and security.

Despite promising developments like ERC-4337, current solutions for gas payments offer only partial relief, still burdening users with the need to hold native tokens and manage underlying complexities.

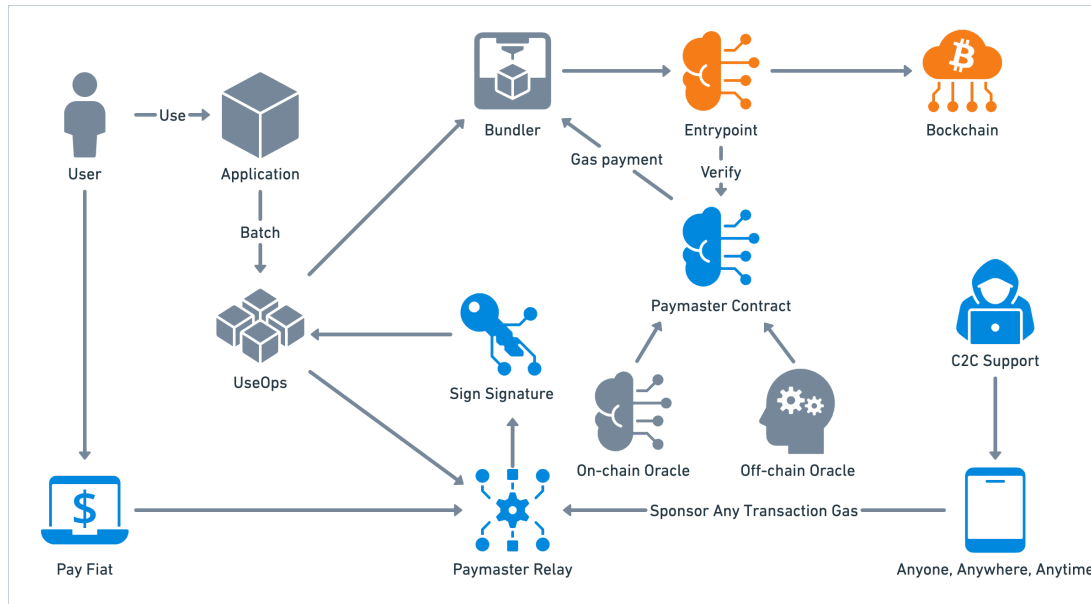


Figure 3: Current Paymaster Solution Flow in Industry

2.2.1 Bad UX: Usability Gap

From both HCI and TAM perspectives[7,8,9,10,11,24,25], current systems exhibit numerous characteristics detrimental to user adoption. They often impose a high cognitive load, demonstrate poor usability across multiple dimensions (detailed in Section 2.3), involve significant direct and indirect costs, necessitate convoluted multi-step processes, offer a super overall experience, and raise security concerns for average users[7]. This poor user experience often stems from the requirement for users to interact with highly abstract concepts lacking clear analogues in their everyday, socially learned experiences, thus failing to align with established mental models[20].

2.2.2 Low Efficiency: Steps and Asset Fragmentation

The multi-step workflow described in 2.1.2 is inherently inefficient. Each step, from CEX onboarding and KYC delays to cross-chain bridge waiting times and transaction confirmation latency, introduces friction and consumes considerable user time and effort. This inefficiency persists even for experienced users, hindering fluid interaction with dApps[32].

The proliferation of diverse blockchain networks (Layer 1s and Layer 2s) necessitates users holding small balances of different native tokens (e.g., ETH on Ethereum mainnet, ETH on Arbitrum, MATIC on Polygon) simply to pay gas fees on each respective chain. This fragmentation increases user overhead, complicates asset management, and adds significant cumulative costs associated with acquiring and managing these disparate gas tokens [18, 28].

2.2.3 Security Risks: Centralized Gas Payment Services (Overview)

The rise of centralized services offering gas sponsorship (Paymasters) or transaction relaying (Bundlers), often associated with ERC-4337 implementations, introduces a new set of risks that potentially undermine blockchain's core principles. These include security vulnerabilities [34,35], the potential for

transaction manipulation or censorship based on the provider's policies or jurisdiction, and the risk of market monopolies leading to inflated costs and reduced innovation [33]. A detailed analysis follows in Section 2.4. It's paradoxical that permissionless accounts, readily created via ECDSA, often necessitate centralized identification and payment methods to acquire gas for initiating decentralized transactions.

2.3 Usability Challenges in Gas Payment: An HCI Perspective

HCI Challenge	The Core Problem	Specific Impact on Users
Ease of Learning	Abstract Concepts & Complex Process: Users must learn numerous new, non-analogous concepts (e.g., Gas, Gwei, addresses) and master a complex, multi-step workflow.	Extremely steep learning curve, making it difficult for new users to quickly learn and navigate the system effectively.
Gulf of Execution	Mismatch Between Intent & Action: A huge gap exists between a user's simple intention (e.g., "buy an NFT") and the required system actions (get native tokens, manage keys, set gas).	Users are confused about how to translate their goals into actions, as the process is unfamiliar and disconnected from traditional financial interactions.
Gulf of Evaluation	Opaque System State: Users struggle to understand transaction costs (volatile gas fees), progress (technical hashes), and failure reasons (cryptic errors), making it hard to assess outcomes.	Users feel uncertain about the results of their actions and are unsure how to proceed after a failure due to a lack of clear, understandable feedback.
Efficiency Issues	Time-Consuming Workflow: The entire process, from KYC and fiat on-ramps to bridging and on-chain confirmation, is plagued by delays, creating a slow and cumbersome experience.	Hinders rapid or spontaneous interactions with dApps, leading to a sluggish and inefficient user experience.
High Error Rate & Low Fault Tolerance	Irreversible & Costly Mistakes: Simple errors like sending to a wrong address, selecting the wrong network, or setting inadequate gas can lead to permanent fund loss, with no "undo" or robust prevention mechanisms.	The stakes are extremely high for users, where a small mistake can be catastrophic. The system is unforgiving of user error.
Memorization Difficulties	Heavy Cognitive Load for Recall: Users are required to securely memorize/store complex seed phrases, distinguish between cryptic addresses, and recall specific procedures for different chains/dApps.	Places a significant burden on user memory, increasing cognitive load and the likelihood of critical errors.
Low User Satisfaction	Poor Overall Experience: The combination of high cognitive load, inefficiency, and the risk of costly errors leads to widespread user frustration and dissatisfaction.	The fundamentally poor usability of gas payments significantly detracts from a positive user experience, regardless of the dApp's utility.
Lack of Supporting Tools	Missing Infrastructure for Developers: The ecosystem lacks standardized, easy-to-integrate tools for developers to build user-friendly gas solutions, making it costly to create smooth experiences.	dApp developers must either rely on complex external wallet UIs or invest heavily in custom solutions, leading to inconsistent user experiences.
High Cognitive Load	Information Overload: Users must process a massive volume of novel technical concepts (Nonce, MEV, etc.) without intuitive metaphors, consuming significant mental effort.	Learning and performing tasks become exceptionally difficult, leaving users feeling mentally exhausted and hindering deeper engagement with the system.
Low Perceived Ease of Use	Negative First Impression: The initial perception is that blockchain systems are inherently complex, expensive, and insecure, failing to map to users' existing interaction patterns.	This perception acts as a major barrier to trial and adoption, deterring potential users before they even experience the underlying dApp's value.

Table 1: Usability Challenges in Gas Payments from an HCI Perspective

2.4 Risk Analysis of Centralized Gas Payment Services

While centralized services aim to simplify gas payments, often leveraging ERC-4337 components like Paymasters, they introduce distinct risks stemming from

their centralized nature, bring new risk to make blockchain be centralized.

2.4.1 Risk Analysis of Centralized Gas Payment Services

Risk Category	Analysis & Mechanism (The "How" and "Why")	Evidence & Specific Examples
Economic & Integration Barriers	Centralized solutions demand that dApp developers integrate proprietary SDKs and accept service agreements. Furthermore, the underlying ERC-4337 smart contract accounts have a higher base gas cost than standard accounts (EOAs), creating an economic disincentive.	• High integration costs for developers. • Inherent gas overhead of ERC-4337 accounts. • Source: [16]
Transaction Manipulation (MEV)	Centralized entities like Bundlers and Paymasters gain a privileged view of the transaction flow. This position enables them to reorder, insert, or delay transactions to extract value from users before transactions are confirmed on-chain.	• Practices: Front-running, sandwich attacks. • Impact: Value is extracted from users' trades at their expense. • Source: [33]
Privacy Leakage	These services become central aggregators of vast amounts of user transaction data. This data, which can be linked to identifiers like IP addresses, creates a single point of failure for user privacy.	• Risks: Data breaches, data sold to third parties, or use for surveillance. • Impact: Reveals user behavior and sensitive financial activity.
Censorship & Regulatory Risk	As centralized entities, these services are subject to jurisdictional laws. They can be compelled to block or censor transactions involving addresses on government sanction lists, undermining the core principle of a permissionless network.	• Example: Blocking transactions to/from addresses on OFAC's sanction list. • Irony: Users must perform KYC/AML on centralized exchanges to fund "permissionless" activities.
Limited Gas Token Support	Paymaster services often restrict which tokens are accepted for gas payments, typically favoring large stablecoins or their own platform tokens. This limits user choice and the utility of a project's native token.	• Impact: Forces users into additional, potentially costly token swaps. • Hindrance: Prevents communities from using their own native tokens for network participation.
Monopoly & Cost Inflation	The market for centralized relayers is already showing significant concentration. This leads to a risk of an oligopoly or monopoly where a few dominant players can control the market, dictate terms, and inflate costs over time.	• Long-term Risks: Increased fees, reduced service quality, and stifled innovation. • Data: Market concentration is shown by Figure 4 (data from BundleBear). • Source: [6]

Table 2: Risk Analysis of Centralized Gas Payment Services

2.4.2 Market Share of Centralized Gas Payment Services

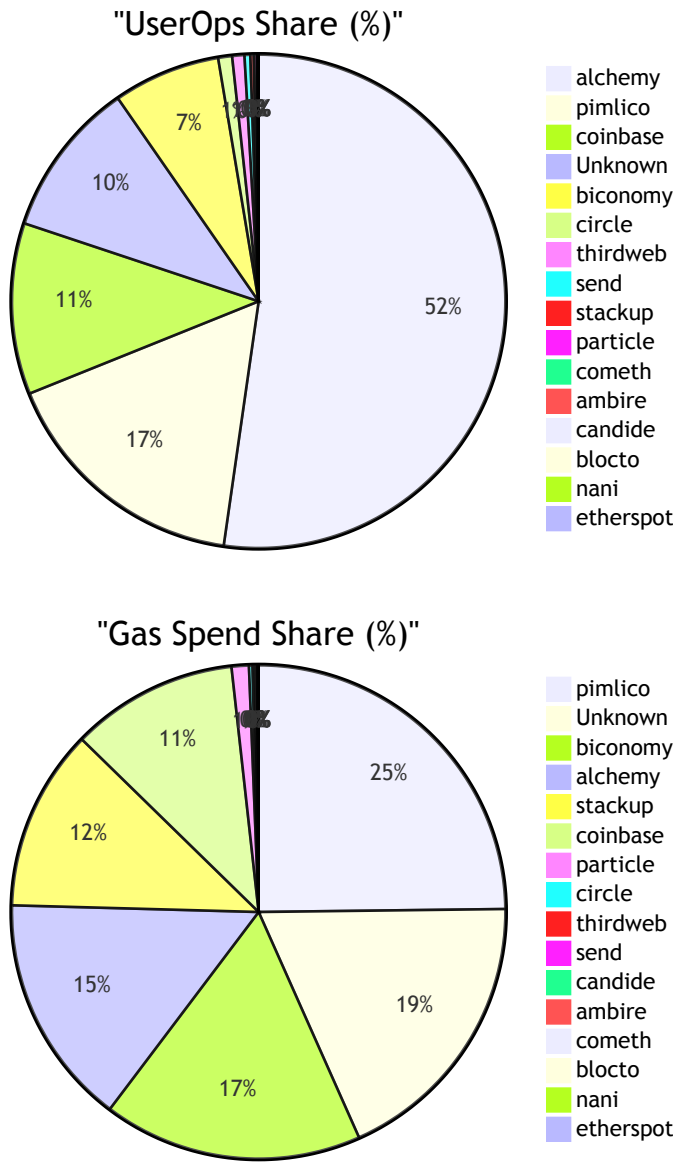


Figure 4: Centralized Paymaster Market Monopoly Graph: UserOps Share and Gas Spend Share, data source: BundleBear

3. The Proposed System Overview: SuperPaymaster

Addressing the multifaceted challenges detailed in Section 2 requires a paradigm shift from centralized or simplistic gas payment solutions towards a decentralized, user-centric, and competitive system, one designed by leveraging familiar user paradigms to abstract technical complexity. This section introduces SuperPaymaster, a novel system designed to achieve this vision. We first outline the critical requirements for such a system, then provide an overview of SuperPaymaster's architecture and actors, delve into its core components including the foundational Standardized Decentralized Service System (SDSS), explain its trust model, and finally, detail its operational workflows and development from both system and user perspectives.

3.1 Core Requirements Table for the SuperPaymaster System

General Requirement	Core Goal (Why this is needed)	Key Components & Mechanisms (How it's achieved)
1. Robustness & Trustworthiness	To build a secure, reliable, and privacy-preserving foundation that users and dApps can depend on, ensuring the integrity of all operations.	• Security: Protect user funds and data with strong authentication (e.g., D2FA) and defense against on-chain attacks. • Privacy: Minimize data exposure and prevent surveillance, potentially using technologies like TEEs. • Availability: Guarantee consistent uptime and fault tolerance through a decentralized network of redundant service nodes.
2. User-Centricity & Economic Viability	To abstract all technical complexity, making gas payments invisible, effortless, and highly affordable for the end-user, thereby creating a Web2-like experience.	• Usability: Lower the learning curve by leveraging familiar mental models like "prepaid cards" or "loyalty points" (via OpenCards/NFTs). • Cost-Effectiveness: Drive down costs with competitive quoting and enable zero/negative cost for users via community points (OpenPNTs). • Efficiency: Ensure swift and streamlined transaction processing to deliver a seamless user experience.
3. Decentralization & Open Competition	To create a fair, open, and permissionless market that prevents censorship and monopolies, ensuring long-term system health, innovation, and community empowerment.	• Competitiveness: Foster a dynamic market among service providers with reputation systems and competitive quoting to ensure fair pricing. • Openness: Build on open-source principles where anyone can participate as a user, developer, or service node operator. • Permissionless: Allow any community to issue its own gas tokens and any node to freely join the network without central approval.

Table 3: Core Requirements Table for the SuperPaymaster System

3.2 Overview of the SuperPaymaster System

SuperPaymaster is proposed as a decentralized gas payment (sponsorship) system built upon the ERC-4337 standard and leveraging a novel Standardized Decentralized Service System (SDSS) architecture. Its core objective is to create an open, competitive, and resilient marketplace for gas sponsorship, addressing the cost, usability, efficiency, and centralization issues prevalent in existing solutions. Key motivations include providing a single, consistent Paymaster address across chains for developer convenience and unifying the staking mechanism for all participating sponsors (LPs/Nodes) to enhance overall system trust and reliability. It facilitates various user-friendly payment models addressing the cost and usability issues mentioned above, all managed within a decentralized framework that utilizes relatable concepts like 'Gas Cards' to simplify user interaction.

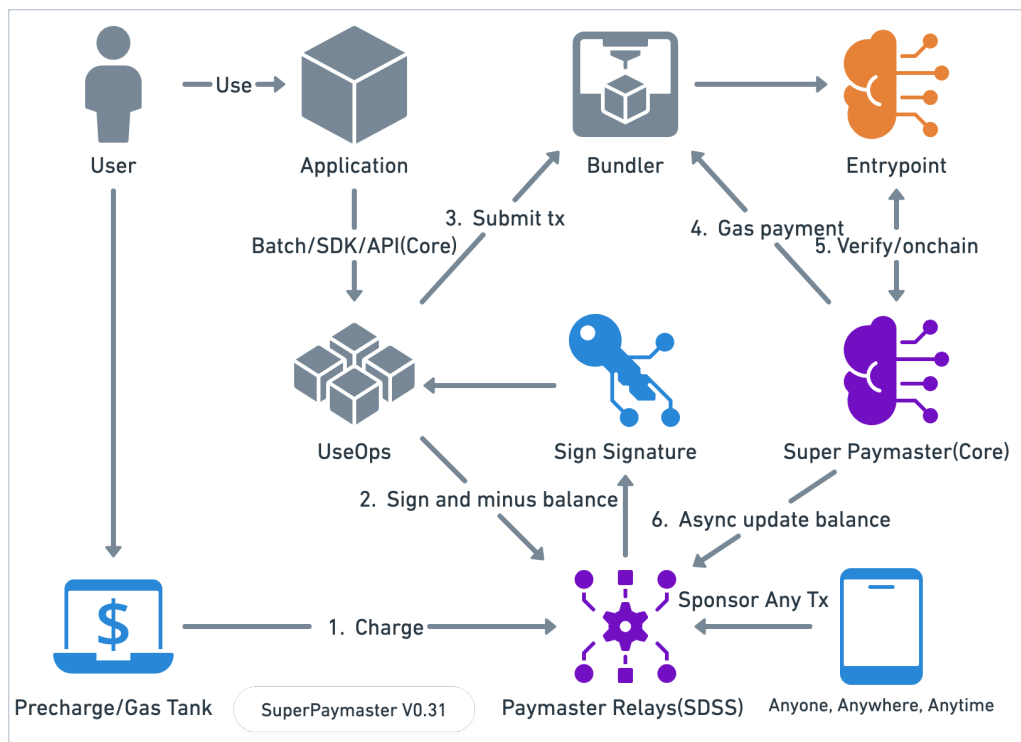


Figure 5: SuperPaymaster System Flow Overview

3.3 Involved Actors and Roles

The SuperPaymaster system involves several key actors based on ERC4337 solution[2]:

End Users Individuals interacting with dApps who require gas payments for their transactions. They benefit from simplified processes, lower costs, and enhanced security via systems like AirAccount.

dApps (Decentralized Applications) Applications integrating SuperPaymaster (via SDSS APIs) to offer seamless gas payment experiences to their users.

Communities Groups or organizations that may issue their own ERC-20 tokens (OpenPNTs) usable for gas payments within the SuperPaymaster network via OpenCards, fostering community engagement.

Node Operators (Paymaster / Gas Sponsors / LPs) Entities running the SuperPaymaster service nodes. They register within the SDSS, stake collateral in the SuperPaymaster contract, listen for gas sponsorship requests, provide quotes, sign UserOperations, and facilitate gas payments. They are incentivized through service fees and reputation gains. Multiple node types (N1, N2, N3 with varying capabilities like TEE) may exist.

Bundlers / RPC Providers Entities responsible for bundling UserOperations (containing Paymaster data) into transactions and submitting them to the blockchain's transaction pool (or directly to block builders under future proposals like RIP-7560[55]).

On-Chain Contracts We use Superpaymaster contract to verify off-chain signature and handle the gas sponsorship and payment. All transaction rely on official EntryPoint contract to verify, launch and guarantee basic security from EIP-4337 team.

Third-Party Swap Services (Optional) Services that may be integrated to facilitate real-time conversion between various ERC-20 tokens and the native gas

token (e.g., ETH) if required by the Paymaster node.

3.4 SDSS (Standardized Decentralized Service System) Overview

SDSS serves as the foundational communication and discovery layer for decentralized services for the SuperPaymaster system. It aims to provide a secure, transparent, and user-friendly architecture for basic decentralized computing services, moving beyond reliance on traditional centralized cloud infrastructure. Its core components facilitate the discovery and interaction with permissionlessly operated service nodes.

(Note: Detailed SDSS components are elaborated in 3.5.2)

3.5 The infrastructure of SuperPaymaster System

SuperPaymaster integrates several key technological and economic components. These components interact to implement the core functionalities and realize the system's design philosophy of mapping complex blockchain operations onto more familiar user experiences.

3.5.1 SuperPaymaster Core

We build SuperPaymaster based on ERC4337, so there are 4 parts:

1. SuperPaymaster contract: stake the ETH and verify the signature, pay the gas sponsorship.
2. SuperPaymaster relay server: handle the user operation and sign a signature before or after deduce your ERC20 token balance.
3. SuperPaymaster ENS API: response to some quota and routing services directly or push to pool timely.
4. SuperPaymaster client/dApps SDK: help developers to initiate the user operation and submit it to bunlder after get the gas sponsorship signature from SuperPaymaster server.

3.5.2 Standardized Decentralized Service System (SDSS) / DePIN

The Standardized Decentralized Service System (SDSS), also conceptualized as a Decentralized Physical Infrastructure Network (DePIN), establishes a core framework for decentralized services. It employs a multi-tiered node architecture, facilitated by a Software Development Kit (SDK), enabling permissionless participation where any entity can operate nodes for self-service or service provision.

1. Node Architecture:

- **N0 (Client Node):** Cross-platform client applications (developed with Tauri+Node.js) that access services provided by N1-N3 nodes.
- **N1 (Foundation Service Node):** Provides static file hosting and Ethereum Name Service (ENS) resolution API services.
- **N2 (Secure Compute Node):** Offers TEE (Trusted Execution Environment) and hardware wallet services, leveraging ARM-based platforms (e.g., Raspberry Pi 5B).
- **N3 (Application Service Node):** Runs containerized services (e.g., Dockerized Supabase instances) for broader application support.

2. Decentralized Service Discovery and Registration: This mechanism underpins the dynamic operation of the SDSS:

- **ENS Registry for Service Discovery:** Leverages the Ethereum Name Service (ENS) for human-readable naming (e.g., `node.ethpaymaster.eth`) and robust service endpoint resolution. Nodes register API endpoints and metadata within ENS records, creating a decentralized alternative to centralized registries.

- **Node Registration Mechanism:** Provides a secure, potentially pseudonymous (via blockchain address) on-chain registry. Nodes register their service capabilities, API endpoints (linked via ENS), and public keys. This process involves staking collateral (e.g., into a SuperPaymaster contract) to ensure accountability and security.
- **Dynamic Routing and Discovery:** Client applications (N0 or dApps) dynamically locate suitable API service nodes by querying the ENS-based registry (e.g., `api.aastar.eth`) or through client-side cached lists. This enables self-maintenance of service records, facilitates failover, and allows for node selection based on criteria such as reputation or network proximity.

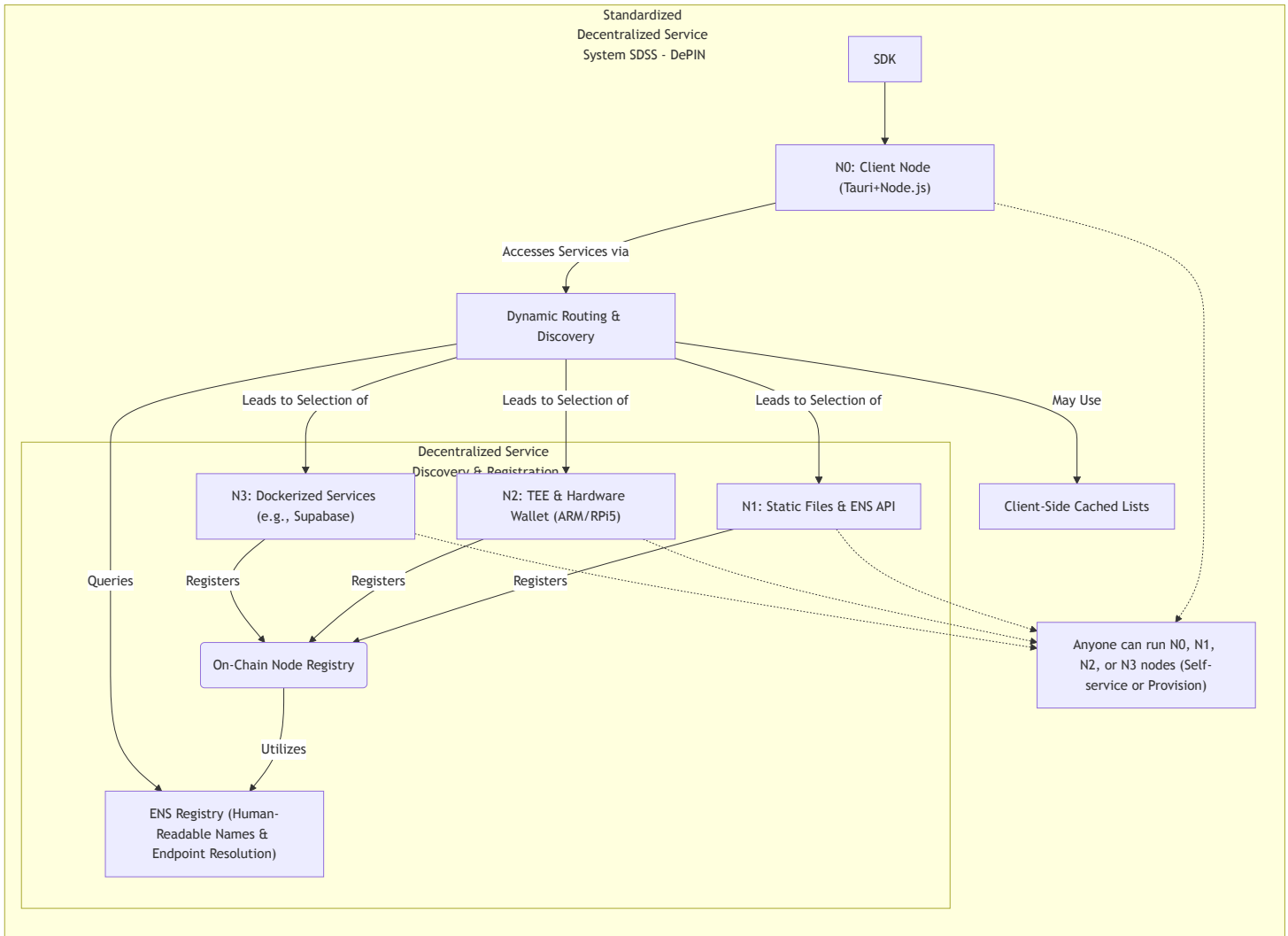


Figure 6: SDSS DePIN Architecture

3.5.3 Competitive Quoting Mechanism

Instead of relying on a single provider's price, dApps can query multiple registered SuperPaymaster nodes (discovered via SDSS) for gas sponsorship quotes for a specific transaction. The dApp can then select the most favorable quote (e.g., lowest cost, highest reputation node, specific ERC20 token), fostering price competition and preventing monopolies. They(dApps) can fetch by a single API from SDSS service to get multiple quotes and access API URLs. All quotes will be updated by SDSS.

3.5.4 Self-Custodial AirAccount Integration: Secure and User-Centric Account Management

This component leverages the AirAccount system to provide robust and user-centric account management. All user operations are securely authorized via AirAccount, employing a dual-factor approach:

1. **Biometric Authentication (D2FA):** Utilizes device-based biometrics (e.g., fingerprint via Secure Enclave/TEE, FIDO2/Passkey standards) for transaction signing. This serves as a decentralized second-factor authentication (D2FA), markedly enhancing security beyond conventional private key models while improving usability through intuitive actions like "Just press fingerprint."
2. **TEE-Secured Private Keys:** Private keys are cryptographically secured within a Trusted Execution Environment (TEE), operating under pre-defined rules to govern their usage.

On-chain, the integrity of every transaction is verified by the smart contract account using BLS12-381 (compliant with EIP-2537) and secp256k1/ECDSA (compliant with ERC4337/EIP-1271) cryptographic algorithms.

Furthermore, this integration confers the inherent advantages of smart contract accounts managed by AirAccount, such as social recovery mechanisms ("moving house"), configurable spending limits, session key management, and the potential for automated will execution features.

3.5.5 Open Community Mode: Flexible Gas Payment & Community Empowerment

The Open Community Mode introduces a flexible framework for on-chain gas payment, empowering communities and users through tangible social and economic constructs. This mode allows communities to operate permissionless Community Nodes, offering guidance and gas sponsorship options to users. It comprises three synergistically integrated components designed to simplify blockchain interactions and reduce cognitive load associated with gas fees.

1. **Permissionless Community Tokens (OpenPNTs):** This mechanism enables any community to issue its own ERC20-compliant tokens (PNTs). Configured SuperPaymaster nodes accept these PNTs for gas payments, fostering and incentivizing community-specific activities. An extension of ERC20, EIP-777, is utilized for efficient token balance deduction.
2. **NFT-based Gas Cards (OpenCards):** Leveraging Soul Bound Tokens (SBT) and Non-Fungible Tokens (NFT) (ERC-721/EIP-6551 compatible), this component implements a 'Gas Card' metaphor based on familiar prepaid card mental models. It automatically recognizes identity and facilitates gas payments, providing cardholders with automatic gas deductions (using PNTs or predefined limits) for a seamless "gasless" experience. This approach mitigates the complexity of blockchain interactions, significantly reducing cognitive load and enhancing Perceived Ease of Use (PEOU).
3. **Task-for-Points Mechanism:** This feature allows users to earn community PNTs by engaging in designated tasks, such as social promotion or content creation. The earned PNTs can subsequently be utilized via OpenCards to cover gas fees, potentially leading to a net-zero or even negative cost for gas transactions.

3.6 Core Gas Sponsorship Workflow (System Perspective)

The following steps outline the typical process flow when a transaction is sponsored via SuperPaymaster:

Step 1: User Account Registration: User registers/logs in via AirAccount (new or bound EOA).

Step 2: Transaction Construction & Encryption: User interacts with a dApp, generating transaction data (UserOperation). AirAccount (using fingerprint/passkey) signs the UserOperation.

Step 3: Gas Price Quoting via SDSS: The dApp (or user's wallet/AirAccount backend) uses SDSS (via ENS lookup) to discover available SuperPaymaster nodes and requests gas sponsorship quotes for the UserOperation.

Step 4: Paymaster Selection & Signature Acquisition: The dApp selects the best quote (based on price, reputation) and requests a Paymaster signature from the chosen node for the UserOperation. The node verifies its ability to sponsor (e.g., checks internal limits, user's OpenCard status/PNT balance if applicable) and provides the signature.

Step 5: Transaction Submission: The dApp/wallet submits the UserOperation (now including the Paymaster signature and data) to a Bundler or directly to an RPC endpoint supporting ERC-4337 (or future AA standards like EIP-7560).

Step 6: On-Chain Verification: The Bundler includes the UserOperation in a bundle transaction sent to the global EntryPoint contract. The EntryPoint verifies the UserOperation, including calling the specified SuperPaymaster contract to validate the Paymaster signature and ensure sufficient deposit/stake exists. AirAccount's signature might also be verified on-chain if needed.

Step 7: Pre-Payment Actions (within Paymaster validatePaymasterUserOp): The SuperPaymaster contract, during validation, checks the sponsoring node's stake and potentially locks funds or deducts pre-paid balance (e.g., from OpenCard associated PNTs). It may need to consult an oracle for real-time token prices if accepting non-native tokens.

Step 8: Contract Execution: If validation succeeds, the EntryPoint executes the core logic of the UserOperation (e.g., token transfer, NFT claim).

Step 9: Post-Payment Actions (within Paymaster postOp): After execution, the EntryPoint calls the SuperPaymaster contract's postOp function. This function handles the actual gas payment reconciliation: paying the Bundler/EntryPoint, potentially refunding excess gas pre-payment to the user/node, updating off-chain account balances, and logging data for reputation calculation.

Step 10: Transaction Confirmation: The Bundler's transaction containing the UserOperation bundle is included in a block and finalized on the blockchain. A transaction hash/receipt is returned.

Step 11: Post-Pay Settlement (Node Level): The sponsoring SuperPaymaster node performs its internal post-transaction accounting, potentially billing users for post-paid services or updating internal ledgers.

A sequence diagram illustrating

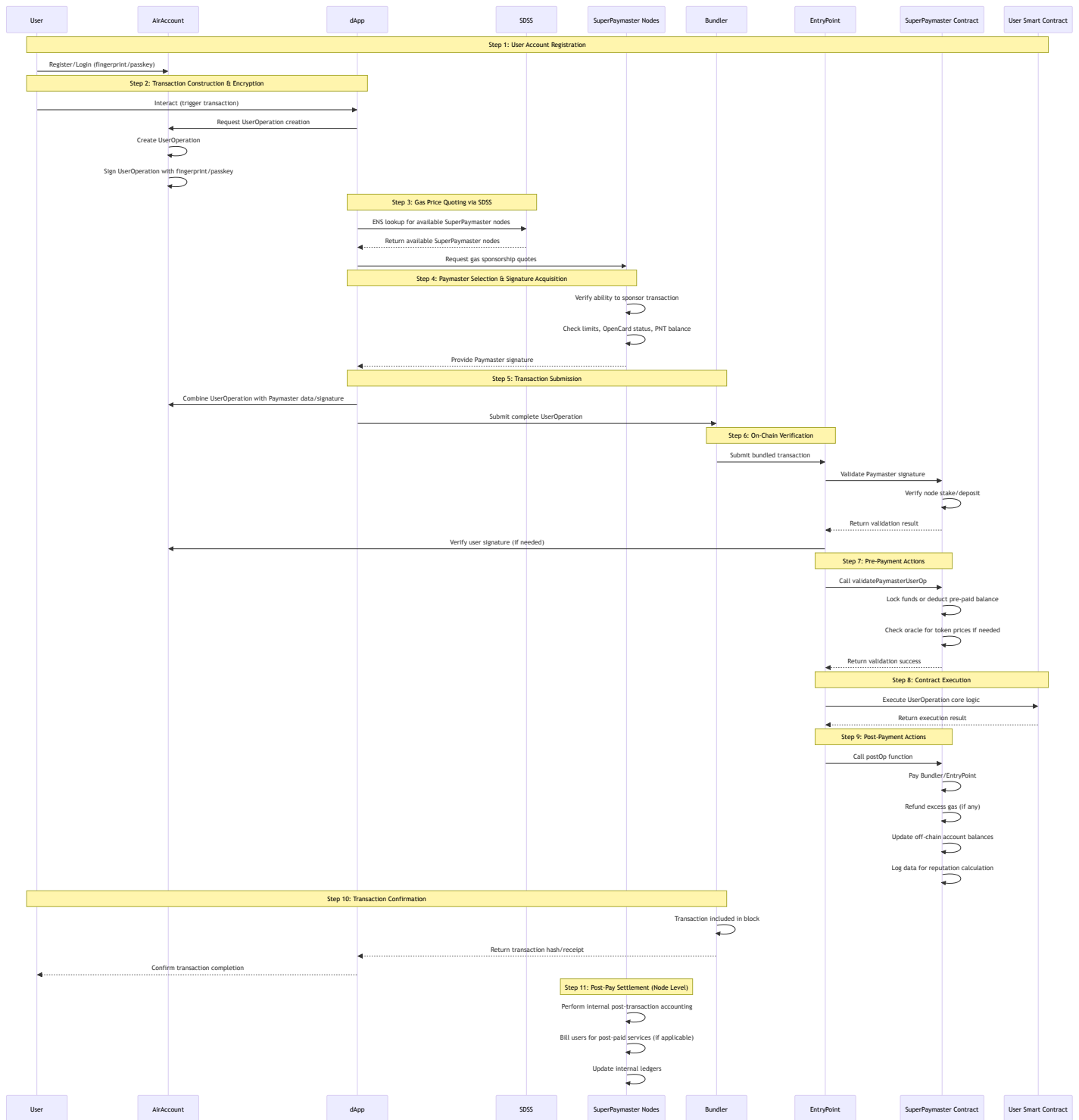


Figure 7: SuperPaymaster Flow

3.7 Typical User Interaction Workflow (User Perspective)

From the end-user's viewpoint, the process is significantly simplified:

Step 1: Accessing Integrated dApp: User visits a dApp that has integrated SuperPaymaster and AirAccount.

Step 2: Account Registration/Login: User logs in or registers using their familiar Web2 method (Google/Email via Passkey/Fingerprint) or by connecting an existing EOA (verified via signature) through the AirAccount interface.

Step 3: Receiving Community NFT/Points (Scenario): Upon joining a community via the dApp, the user might automatically receive an OpenCard NFT representing a

gas allowance or initial OpenPNTs balance.

Step 4: Earning Points (Optional): User participates in community activities (e.g., sharing content, completing tasks) and earns OpenPNTs credited to their account/OpenCard.

Step 5: Initiating Transaction: User interacts with the dApp as intended (e.g., clicks "Mint NFT," "Transfer Tokens," "Buy Item").

Step 6: Seamless Gas Payment: The underlying SuperPaymaster system handles the gas payment. If the user holds a valid OpenCard with sufficient PNTs/allowance, the gas fee is automatically deducted or covered without requiring user confirmation or interaction with native tokens. The experience feels "gasless."

Step 7: Viewing Transaction Records: User receives confirmation of the successful transaction within the dApp, similar to a Web2 purchase confirmation. They can view details like transaction ID and status.

Step 8: Checking Balances and History: User can check their OpenPNTs balance, OpenCard status, and transaction history (including gas cost savings estimations) within their AirAccount interface or the dApp.

Sequence Diagram

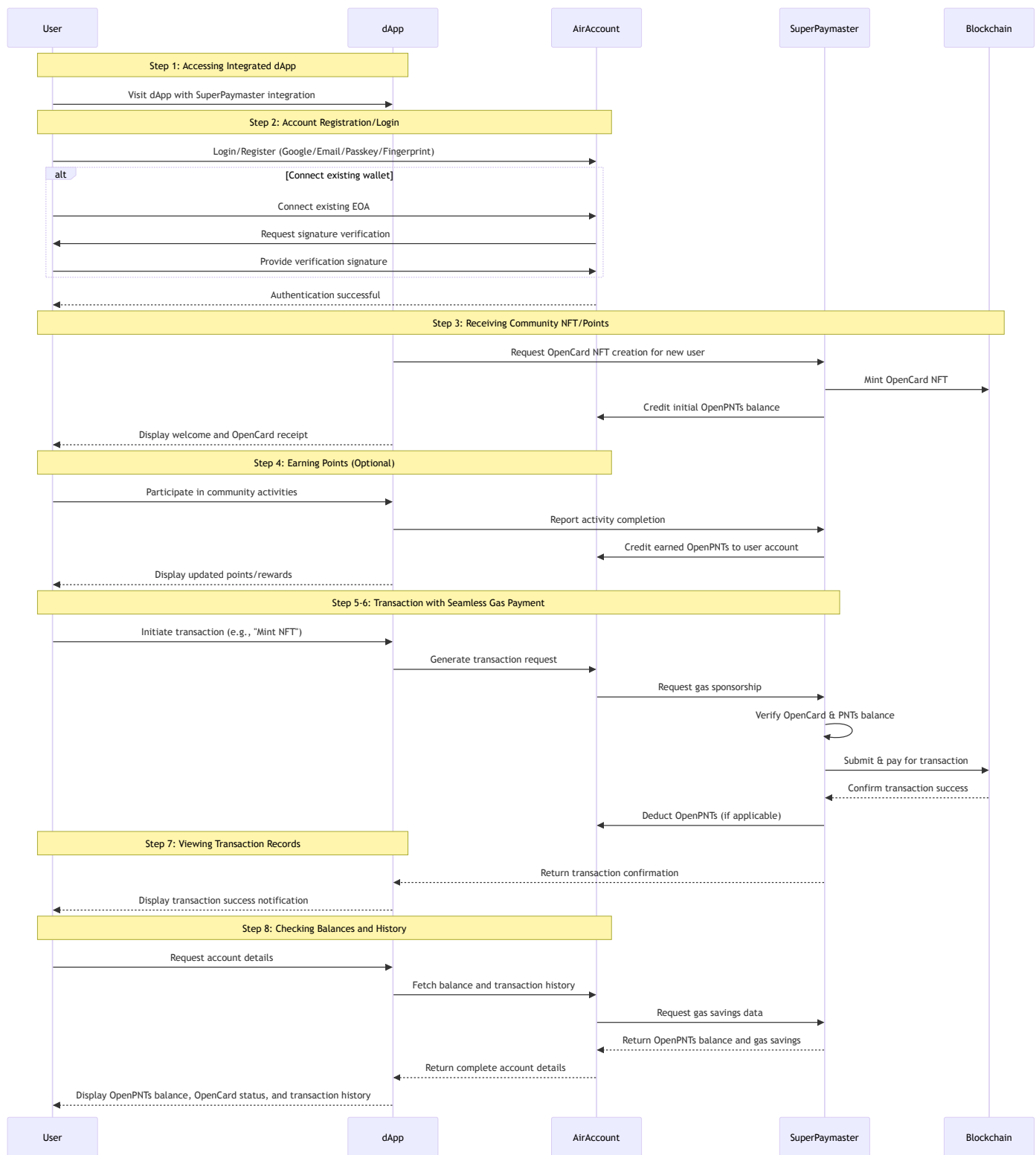


Figure 8: User Interaction Workflow

4. Implementation (Proof of Concept - PoC)

This section details the Proof of Concept (PoC) implementation of the SuperPaymaster platform, covering smart contract development, the Standardized Decentralized Service System (SDSS) backend, node management, and user interface construction. Technological choices focused on enabling core functionalities—decentralized gas sponsorship, competitive quoting, enhanced user experience, and meeting security and interoperability requirements.

The SuperPaymaster PoC utilized the following tools and frameworks: Smart Contracts: For immutable consensus, we use Solidity[37] within the Foundry[38], a high-performance contract development framework implemented the core SuperPaymaster contract and auxiliary contracts for ENS resolution, OpenPNTs (ERC-20), OpenCards (ERC-721), and node registration which based on open source infrastructure. User Interfaces: Use popular and easy-to-use JavaScript/TypeScript framework, Next.js Next.js[39] (React[40], Node.js[41]) built web frontends. Desktop Clients: Tauri [42] packaged cross-platform clients (Windows, MacOS, Linux, iOS, Android, Web) supporting N0 user nodes and N1 service nodes. Backend Services: High concurrency support, Go Golang [43] and Rust [44] developed backend/node on libp2p; Rust targeted performance-critical components and potential TEE-based services (N2 nodes). Deployment & Infrastructure: Docker Docker [45] provided containerization. A customized Supabase [46] instance handled BaaS, database, and storage functionalities (N3 nodes). Account Management: The permissionless AirAccount API [47] managed user account lifecycles, security, and authentication. Hardware Testing on Decentralized Physical Infrastructure Networks(DePIN)[48]. We use high performance and Affordable for everyone like NXP i MX95(or TI AM64xx) served as representative compute nodes in production, use Raspberry Pi 5B 16G for developing and testing[78].

4.1 System Setup and Configuration

We need to setup AirAccount, SuperPaymaster Nodes Configuration, to set interaction config with decentralized account supporting gas sponsorship, and a basic config for OpenPNTs, OpenCards and more parameters to pay your gas seamlessly. Also we need create cross-chain CometENS API name for node registry to get decentralized invoking. Anyone can run a node to act as a Paymaster Service Provider with their Secp256k1 key staked and registered on-chain with their own ERC20 gas token.

```

{
  "name": "AAstar SuperPaymaster Config Demo",
  "description": "A decentralized gas sponsor provider node",
  "image": "https://aastar.io/superpaymaster.png",
  "url": "https://aastar.io/superpaymaster",
  "ens": "paymaster.aastar.eth",
  "address": "0x1234567890123456789012345678901234567890",
  "stake": {
    "eth": "1000",
    "aastar": "1000",
    "promise": {
      "duration": "30d",
      "amount": "1000",
      "item": "url/ipfs",
      "token-accept": {
        "eth": "0x0000000000000000000000000000000000000000",
        "astPNTs": "0x1234567890123456789012345678901234567890",
        "USDT": "0x1234567890123456789012345678901234567890",
        "USDC": "0x1234567890123456789012345678901234567890",
        "DAI": "0x1234567890123456789012345678901234567890",
        "WETH": "0x1234567890123456789012345678901234567890"
      },
      "price": {
        "eth": "30",
        "astPNTs": "30",
        "USDT": "30",
        "USDC": "30",
        "DAI": "30",
        "WETH": "30"
      }
    },
  },
  "openpnts": {
    "factory": "0x1234567890123456789012345678901234567890",
    "PNTs": "0x1234567890123456789012345678901234567890",
    "ratio": "ratio.aastar.eth",
    "symbol": "astPNTs"
  },
  "opencards": {
    "factory": "0x1234567890123456789012345678901234567890",
    "nft": "0x1234567890123456789012345678901234567890",
    "ratio": "ratio.aastar.eth",
    "symbol": "astCards"
  },
  "Paymaster config": {
    "token-accept": [{
      "symbol": "astPNTs",
      "address": "0x1234567890123456789012345678901234567890",
      "price": "30"
    }, {
      "symbol": "xPNTs",
      "address": "0x0000000000000000000000000000000000000000",
      "price": "20"
    }],
    "limitation": {
      "daily": "1000",
      "single": "1 ETH"
    }
  }
}

```

4.2 Smart Contract Design and Development

Smart contract is the key part of the system, through mutable on-chain code, to ensure verification of gas payment signatures, payment of gas, deduction of reasonable PNTs, allocation of PNTs income, calculation of reputation (success rate) and Slash etc. We only introduce core ability of SuperPaymaster contract, more details can be found in appendix.

1. Stake: Sub-account stake management for security and gas sponsor
2. Verify and Pay: Sub-account signature verification, payment, record and balance maintenance
3. Post Processing: Transaction success post processing: reputation increase
4. Compensation: Asynchronous transaction status compensation: failed and successful re-check, proof submission and reputation modification (off-chain, call on-chain method)

4.2.1 SuperPaymaster Contract Flow

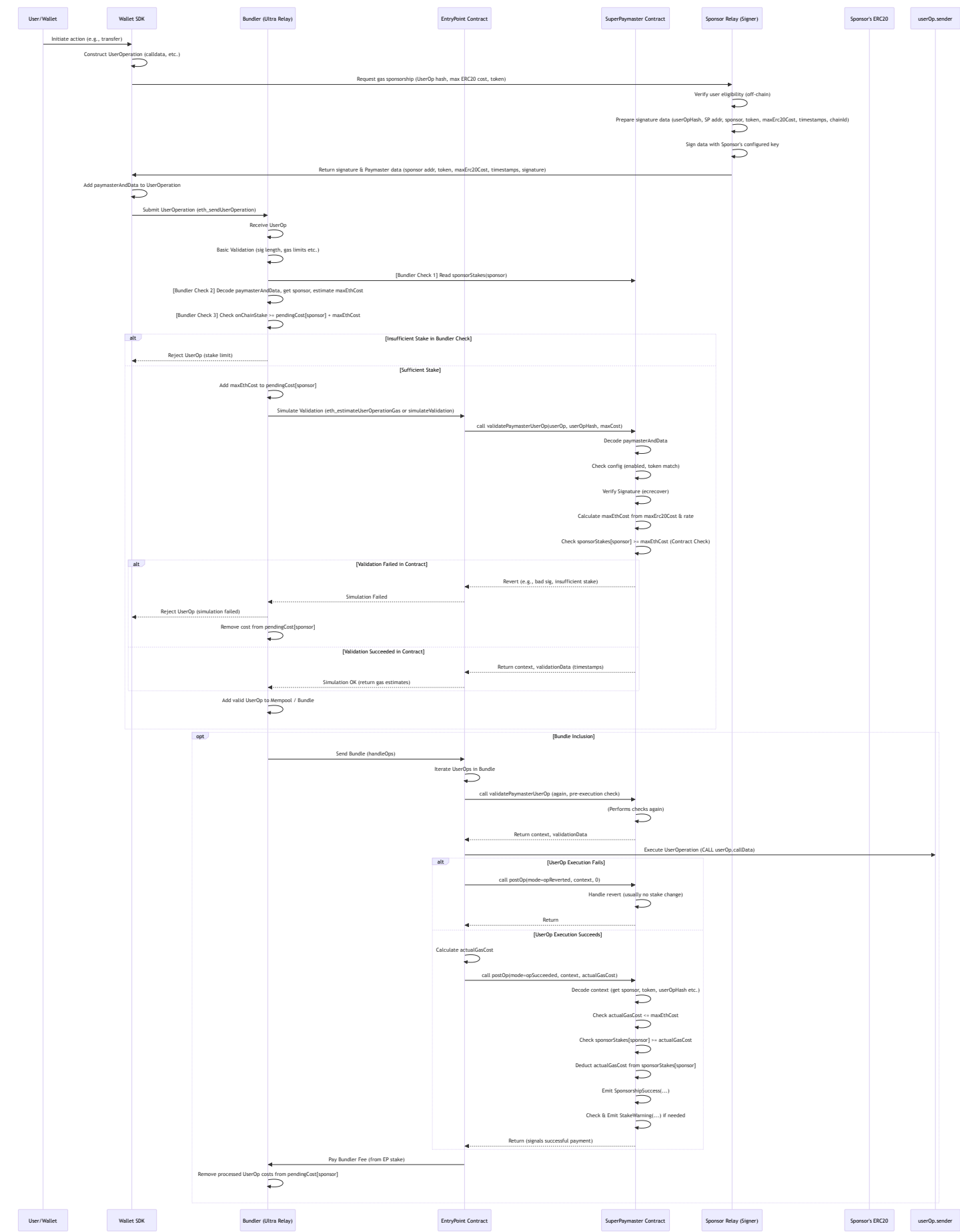


Figure 9: SuperPaymaster Contract Work Flow

4.2.2 SuperPaymaster Contract Main Functions

The smart contract has two main functions: stake and verifyAndPay, more details can be found in [56]. We build this contract based on ERC4337 and other Open-source projects: Pimlico singleton paymaster[57] and ZeroDev bundler[58]. We add sub-account stake module to permit permissionless staker to run their own paymaster for supporting their own ERC20 gas tokens. We improve verify function to avoid over-use risk with on-chain contract and off-chain bundler. Also we build SuperPaymaster relay server framework to support dApps to query gas price and pay gas with different ERC20 supporting. Stake manager contract provide a stake reputation and ETH payment guarantee for every ERC20 gas payment.

```

// SuperPaymaster.sol main function1: Stake manager
/*    SPONSOR MANAGEMENT    */

/**
 * @notice Set the withdrawal delay period
 * @param _withdrawalDelay New delay period in seconds
 */
function setWithdrawalDelay(uint256 _withdrawalDelay) external onlyAdminOrManager {
    require(_withdrawalDelay > 0, "SuperPaymaster: withdrawal delay must be positive");
    withdrawalDelay = _withdrawalDelay;
}

/**
 * @inheritdoc ISuperPaymaster
 */
function registerSponsor(address sponsor) external override onlyAdminOrManager {
    require(!isSponsor[sponsor], "SuperPaymaster: sponsor already registered");
    isSponsor[sponsor] = true;

    // Initialize with default config (owner = sponsor itself)
    sponsorConfigs[sponsor] = SponsorConfig({
        owner: sponsor,
        token: address(0),
        exchangeRate: 0,
        warningThreshold: 0,
        isEnabled: false,
        signer: address(0)
    });

    emit SponsorRegistered(sponsor);
}

/**
 * @inheritdoc ISuperPaymaster
 */
function setSponsorConfig(
    address token,
    uint256 exchangeRate,
    uint256 warningThreshold,
    address signer
) external override {
    address sponsor = msg.sender;
    require(isSponsor[sponsor], "SuperPaymaster: not a sponsor");
    require(msg.sender == sponsorConfigs[sponsor].owner, "SuperPaymaster: only sponsor can modify settings");
    require(token != address(0), "SuperPaymaster: invalid token address");
    require(signer != address(0), "SuperPaymaster: invalid signer address");

    SponsorConfig storage config = sponsorConfigs[sponsor];
    config.token = token;
    config.exchangeRate = exchangeRate;
    config.warningThreshold = warningThreshold;
    config.signer = signer;

    emit SponsorConfigSet(sponsor, token, exchangeRate, warningThreshold, signer);
}

```

verifyAndPay provides a security guarantee for dApps to pay gas with different ERC20 tokens under their credit stake amount.


```

// SuperPaymaster.sol main function2: verifyAndPay
function validateSponsorUserOp(
    PackedUserOperation calldata userOp,
    bytes32 userOpHash,
    uint256 /*requiredPreFund*/,
    bool allowAllBundlers,
    bytes calldata paymasterConfig
) internal returns (bytes memory context, uint256 validationData) {
    // check bundler authorization
    if (!allowAllBundlers && !isBundlerAllowed(tx.origin)) {
        revert BundlerNotAllowed(tx.origin);
    }

    // check replay attack
    require(!processedOps[userOpHash], "SuperPaymaster: operation hash already processed");

    // parse sponsor data
    (
        address sponsor,
        address token,
        uint256 maxErc20Cost,
        uint48 validUntil,
        uint48 validAfter,
        bytes calldata signature
    ) = _parseSponsorConfig(paymasterConfig);

    // verify sponsor is valid
    require(isSponsor[sponsor], "SuperPaymaster: invalid sponsor");
    require(sponsorConfigs[sponsor].isEnabled, "SuperPaymaster: sponsor not enabled");

    // verify token is matched
    require(token == sponsorConfigs[sponsor].token, "SuperPaymaster: token mismatch");

    // get sponsor signer
    address signer = sponsorConfigs[sponsor].signer;

    // create message hash for signature verification
    bytes32 hash = _getSponsorHash(userOp, userOpHash, sponsor, token, maxErc20Cost, validUntil, validAfter);

    // verify signature
    (bytes32 r, bytes32 s, uint8 v) = _extractSignature(signature);
    address recoveredSigner = ecrecover(hash, v, r, s);

    // check signature is valid
    if (recoveredSigner != signer) {
        revert("SuperPaymaster: invalid sponsor signature");
    }

    // calculate max ETH cost
    uint256 exchangeRate = sponsorConfigs[sponsor].exchangeRate;
    require(exchangeRate > 0, "SuperPaymaster: invalid exchange rate");

    // calculate maxEthCost: (maxErc20Cost * 1 ether) / exchangeRate
    uint256 maxEthCost = (maxErc20Cost * 1 ether) / exchangeRate;

    // get sponsor stake
    EnhancedSponsorStake storage stake = sponsorStakes[sponsor];

    // ensure sponsor has enough stake
    require(
        stake.stakedAmount >= maxEthCost,
        "SuperPaymaster: insufficient sponsor stake"
    );

    // lock this operation's funds

```

```

    if (stake.userOpLocks[userOpHash] == 0) {
        stake.lockedAmount += maxEthCost;
        stake.userOpLocks[userOpHash] = maxEthCost;
        emit StakeLocked(sponsor, userOpHash, maxEthCost);
    }

    // pack validation data (signature validity and timestamp)
    validationData = _packValidationData(false, validUntil, validAfter);

    // encode context for postOp
    context = abi.encode(sponsor, token, maxEthCost, maxErc20Cost, userOpHash);

    emit UserOperationSponsored(userOpHash, userOp.getSender(), SPONSOR_MODE, token, maxErc20Cost, maxEthCost);

    return (context, validationData);
}

```

4.2.3 CometENS System Smart Contract

ENS is integral to SDSS, providing decentralized registration and discovery for services. Our CometENS solution utilizes ENS to allow stakers to register unique cross-chain service identifiers and store structured on-chain data (JSON via Text records). This supports dynamic discovery and configuration. Crucially, ENS also improves usability by mapping addresses to human-readable names—analogueous to OpenCards' payment metaphor—leveraging users' understanding of domain systems to simplify interaction with blockchain identifiers.

A typical JSON config in ENS text record:

```

baseUrl = "https://api.aastar.io";
baseENS = "api.aastar.eth";
nodeName = "XXXDAO";
serviceList = [{
  name: "getAPIList",
  description: "return API list",
  method: "GET",
  path: "/api/v1/getAPIList",
  params: [],
  response: [{
    name: "name",
    type: "string",
    required: true,
  }],
}, {
  name: "getAcceptERC20List",
  description: "return Accept ERC20 list",
  method: "GET",
  path: "/api/v1/getAcceptERC20List",
  params: [],
  response: [{
    symbol: "astPNTs",
    address: "0x0000000000000000000000000000000000000000000000000000000000000000",
    price: "30",
  }],
}, {
  name: "getSignature",
  description: "return Tx signature",
  method: "POST",
  path: "/api/v1/getSignature",
  params: [{
    name: "tx_data",
    type: "json",
    required: true,
  }],
  response: [{
    name: "tx_data signature",
    type: "string",
    required: true,
  }],
}];

```

Besides setTextRecord, CometENS also provides normal wallet address setName and resolveName, setName to set your own node address, setAvatar to set your own wallet avatar, setContenthash to help node to have a readme web page in IPFS or other hash address.

Total Entity Relation Graph:

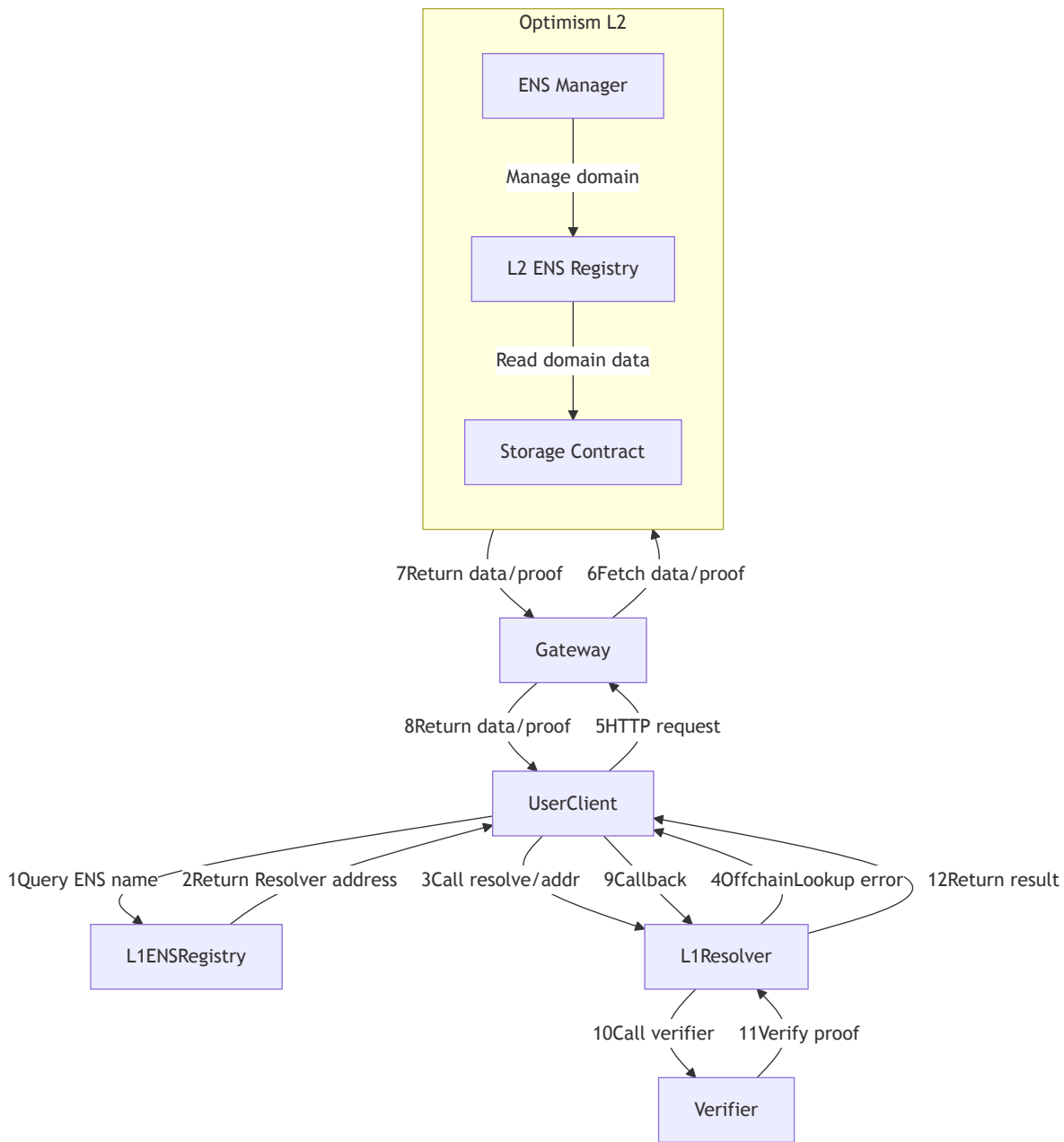


Figure 10: CometENS Flow

We need deploy a series of contracts in Optimism Layer2 and Ethereum mainnet.

Contracts relation graph:

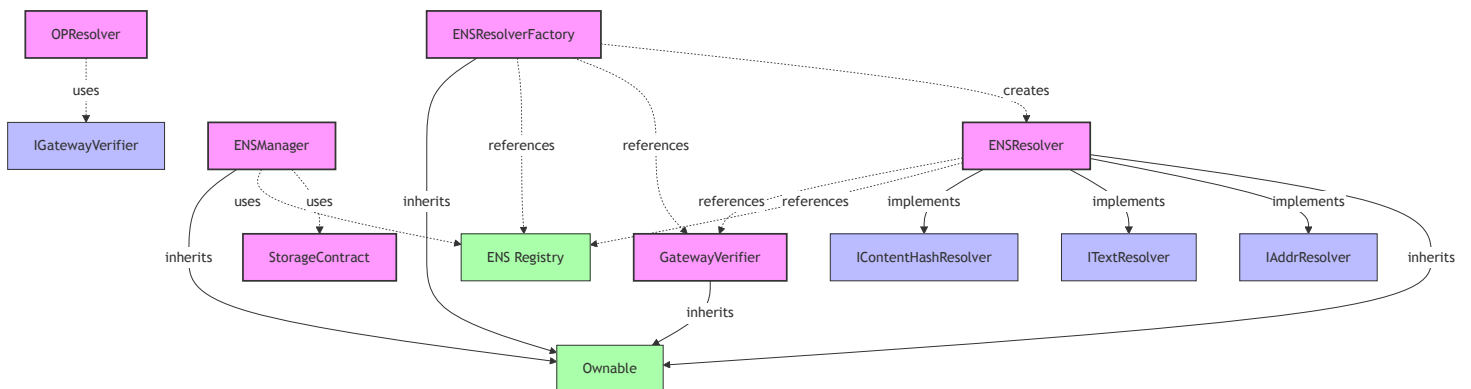


Figure 11: Contracts Relation Graph

4.2.4 OpenPNTs/OpenCards

We need a new token standard to support ERC20 gas token payment, named OpenPNTs.

We also need a new NFT standard to support seamlessly gas payment, named OpenCards. The technical implementation of the OpenPNTs/OpenCards contract serves to realize the 'Gas Card and Points' metaphor, designed to abstract the complexities of gas payment for the end-user.

4.3 AirAccount Integration

SuperPaymaster need to integrate with AirAccount by API to support a totally user case flow for any transaction gas sponsor scenario. We assume any dApp(decentralized application) want to get a seamless and gasless transaction. We should provide the core APIs to support this flow:

1. after Next.js Application initiation.
2. install AirAccount SDK and SuperPaymaster SDK (both are in AASDK)

```
pnpm install aasdk @aasdk/superpaymaster
or
pnpm install aasdk
```

3. Initiate an AirAccount account, used to save PNTs and test transactions.

```
aasdk airaccount create //only support Email binding in command line
or use web page action
```

4. Create a useroperation data(transaction data)

```
{
  "sender": "0x...",           // User's account address (smart contract account)
  "nonce": "0x...",           // Account's nonce for replay protection
  "initCode": "0x...",         // Optional: Code to deploy the account if it doesn't exist
  "callData": "0x...",         // The transaction payload to execute
  "callGasLimit": "0x...",     // Gas limit for executing callData
  "verificationGasLimit": "0x...", // Gas limit for account validation
  "preVerificationGas": "0x...", // Gas to cover off-chain costs (e.g., data gas)
  "maxFeePerGas": "0x...",     // Maximum fee per gas for inclusion
  "maxPriorityFeePerGas": "0x...", // Maximum priority fee per gas
  "paymasterAndData": "0x...", // Paymaster address and data for gas sponsorship
  "signature": "0x..."       // Signature authorizing the UserOperation
}
Gas payment information:
{
  paymentType: "ERC20/OpenCards/ETH" // ERC20 token and OpenCards credential
  // we support 3 types: ERC20 token(in your account balance), OpenCards(binding with a NFT card, use this balance) , and default I
  gasToken: "0x123..." // gas token address
  price: "30" // price per gas for gWei with one ERC20 token
}
```

5. Follow the flow above, send to relay(provides bundler and paymaster service all in one)

4.4 Backend Service Implementation

Node Registry

All backend services must first be registered as nodes. We use Next.js to interact with on-chain contract.

1. Generate a node public/private key pair.
2. Call the node registration contract (NodeRegistry) ABI to register the node.
3. Approving, staking and some form of authorization are required.

- Then the node will can provide some decentralized services like paymaster sponsor services.
- Follow the deployment documents to run a paymaster relay in your node.

Node Registry Information

English ▾

[Register New Node](#) [Back to Home](#)

Contract Address: [0xE1A3B41be95Ff379DBDfd194680d26b5d8786462](#)

Status: Status: **Connected**

Minimum Stake Amount: **1000.0** tokens

Total Nodes: **12**

Query Node Info

Registered Nodes

Node Address 0xe24b6f321B0140716a2b671ed0D983bb64E7DaFA IP/Domain node.cmuba.org API Services 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13 Registration Time 2/2/2025, 12:59:12 PM Active	Node Address 0x26f9fEE42A123083c0De5F213767eE8c25682859 IP/Domain node.cmuba.org API Services 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13 Registration Time 2/2/2025, 1:20:34 PM Active	Node Address 0x202399F0854E99E7458247845eCAad05c8368b50 IP/Domain localhost API Services 1, 2, 3, 4, 5 Registration Time 2/2/2025, 5:14:28 PM Active
Node Address 0x80c0df0e384BBf1735b21FAC3E244796b98A74C1 IP/Domain	Node Address 0x62fbe25960a634b0f6849Ead041166e6bF29e83F IP/Domain	Node Address 0xF0d8c03a0F35e159Cbe4Ff107527EbC9b916B3A5 IP/Domain

Figure 12: Node Registry Flow

SuperPaymaster Relay Server

We create a all-in-one relay server: provide paymaster signature and bundler service. So as the flow above mentioned, the dApp can create the useroperation and then send to the relay to launch the transaction in one time API invoking. We develop the relay based on ZeroDev's ultra relay [58] and Pimlico's Alto bundler [59] source code.

Before a dApp invoking the relay server, you must use ethers.js or wagmi to get a random relay server API URI:

- access website: paymaster.aastar.io(superpaymaster.aastar.io) or parse ENS: paymaster.aastar.eth(superpaymaster.aastar.eth) , get a paymaster collection like this:

```

{
  "paymasters": [
    {
      "address": "0x...",
      "name": "Paymaster 1",
      "ensName": "Paymaster1.0gas.aastar.eth",
      "description": "Description 1",
      "URL": "https://paymaster1.com/api",
      "price": { //gwei price for token
        "eth": "30",
        "astPNTs": "30",
        "USDT": "30",
        "USDC": "30",
        "DAI": "30",
        "WETH": "30"
      }
    },
    {
      "address": "0x...",
      "name": "Paymaster 2",
      "ensName": "Paymaster2.0gas.aastar.eth",
      "description": "Description 2",
      "URL": "https://paymaster2.com/api",
      "price": {
        "eth": "30",
        "astPNTs": "30",
        "USDT": "30",
        "USDC": "30",
        "DAI": "30",
        "WETH": "30"
      }
    }
  ]
}

```

The website will refresh every 10 minutes from on-chain text record. The ENS name is realtime(depends on the specific layer2 sync cycle with layer1). The URL is the API endpoint for the paymaster service. You can remember the paymaster1.0gas.aastar.eth and paymaster2.0gas.aastar.eth to get refresh changes. 2. Determine the fittable paymaster you want to use based on your token type and affordable price, get the URL to invoke. 3. Send the useroperation to the relay server. 4. The relay server will return the paymaster signature and original transaction data.

Let's know the basic flow:

1. whoRU: SuperPaymaster relay server will return the node identity: address, ENS name, public key and ERC20 gas token support list with price.
2. _isRegistered: SuperPaymaster contract can verify whether the node is registered and get the stake amount and reputation.
3. getSignature: SuperPaymaster relay server receive dApps useroperation and gas payment config, then return the paymaster signature and original transaction data.
4. _verifySecondSignature: It is a validator module of relay server to operate the dApps request, validate transaction data and user's finger-print signature.
5. _signPaymasterAndData: SuperPaymaster relay server sign signature after verification.
6. _signTEE, it is a extend function of validator with a ARM chips.
7. _payERC20Gas: SuperPaymaster will check the OpenCard NFT ERC20 balance or account PNTs balance and deduct PNTs firstly.

8. `_postPayment`: if sponsor successfully, refund PNTs and calculate the node reputation.
9. `_simulateTx`: try to simulate the transaction data verify and submit, if passed, send to RPC. It is a bundler module.

4.6 SDSS(Standard Decentralized Service System) Implementation Details

Every application require a backend service or cloud computing service. We create SDSS as a decentralized computing service system that provides a rain computing service for decentralized applications. We have design and evaluate the SDSS system, for the initial version, we will provide a simple service package.

1. A Tauri based cross platform client SDK, anyone can develop their own dApp with this framework with Rust.
2. A permissionless node registry to stake and run your own rain computing node, include CometENS and node management contract with Node.js.
3. A recommendation hardware for developing mode: Raspberry 5B based TEE+hardware wallet node, include hardware wallet and TEE security module in Rust.
4. A open source infura service package:
 - i. Bundler: Ultra Relay +(contract and SDK in Node.js)
 - ii. Paymaster: SuperPaymaster (contract, relay and SDK in Node.js and Go)
 - iii. Account: AirAccount (contract and SDK in Node.js, Rust and Go)
 - iv. Validator: BLS+TEE(service and SDK in Node.js)
 - v. BaaS: Supabase(SDK in Go and Node.js)
 - vi. User dApps Demo: COS72 (SDK and demo in Node.js and Rust)
 - vii. Node management: COS72 Community Panel(UI application) in Node.js and Rust.
 - viii. Docker image: AASstar All regions are categorized by CPU type into ARM, x86, and others. Each region is divided into two parts: User and Community. These two parts have overlapping areas where users with spare computing power act as community computing nodes. User section: Tauri cross-platform client, providing an interactive interface for dApps, can run on any device. Community section: For non-ARM regions, users run Docker services including Bundler, Paymaster, Account, Validator, BaaS (Supabase), and other services. For ARM regions, added TEE hardware wallet + signature service + privacy computing service. Community section: A community panel provides a management portal for configuring all backend services. Hardware DePIN recommendation: Raspberry Pi 5B 16G(for developement), running Docker + TEE is sufficient to serve a small community. We have a backend system module structure graph:

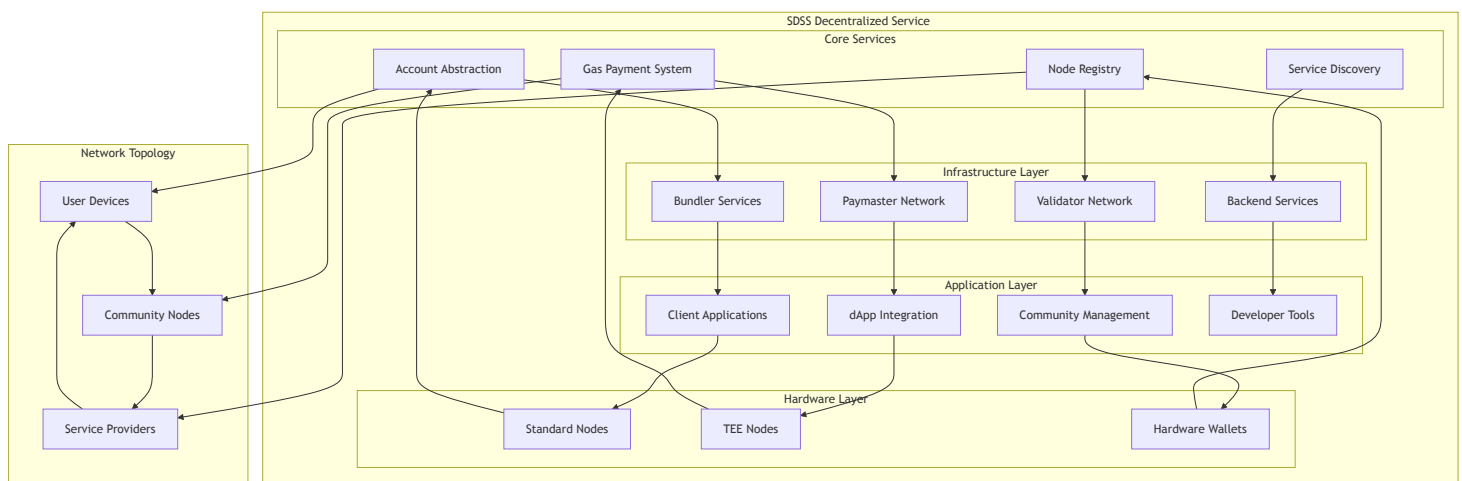


Figure 13: Backend System Module Structure Graph

5. Discussion

This section discusses how the proposed SuperPaymaster system, leveraging Account Abstraction (ERC-4337) and the Standardized Decentralized Service System (SDSS), directly addresses the usability challenges (complexity, cost, efficiency) and the risks associated with centralized gas payment solutions detailed in Section 2.

5.1 Countermeasures Against Usability Challenges

The SuperPaymaster system's design, particularly its strategic application of familiar metaphors like the 'Gas Card,' grounded in principles of social learning and Human-Computer Interaction (HCI) [7,8,9,10,11,20], provides direct countermeasures to the significant usability barriers identified in Section 2.3.

5.1.1 NFT Cards (OpenCards)

for Seamless Payment: A primary usability hurdle is the cognitive load associated with understanding and managing abstract Gas concepts. The 'Gas Card' metaphor, implemented via OpenCards (NFTs/STs) holding OpenPNTs balances, directly tackles this by mapping the gas payment process onto the user's familiar mental model of prepaid or loyalty cards. [Optional: Reference PoC evaluation results here if available] The PoC implementation demonstrates how this reduces the perceived complexity and enhances the Perceived Ease of Use (PEOU) [24]. By automating the gas payment deduction from the card's balance via the Paymaster, the user interaction becomes seamless, approaching a "gasless" experience and significantly lowering the Gulf of Execution [36].

5.1.2 Competitive Quoting

for Cost Reduction: The high and volatile cost of gas presents a major usability challenge. SuperPaymaster addresses this through its competitive quoting mechanism facilitated by SDSS. Instead of being subject to a single provider's rate or needing to manually optimize gas prices, dApps (on behalf of users) can solicit quotes from multiple decentralized Paymaster nodes. This introduces market competition, driving down sponsorship costs and providing users with more predictable and affordable transaction fees, thereby improving user satisfaction and reducing financial barriers.

5.1.3 Community Tokens (OpenPNTs)

for Low/Negative Cost: The necessity of acquiring native tokens (Section 2.2.2) adds cost and complexity. The OpenPNTs framework allows communities to issue their own tokens acceptable for gas payments within the SuperPaymaster network. Combined with mechanisms for users to earn PNTs (e.g., Task-for-Points), this enables pathways to zero or even negative net gas costs for active community participants, fundamentally altering the economic aspect of the usability challenge.

5.1.4 Reputation Mechanism

for Success Rate Guarantee: Transaction failures due to poorly configured gas or unreliable intermediaries contribute to user frustration (Section 2.3.5). The SuperPaymaster Trust Model incorporates an on-chain Reputation Mechanism (Section 3.6.2) that tracks node performance (success rate, uptime, stake). By allowing dApps/users to select nodes based on reputation, the system incentivizes reliability and increases the likelihood of successful transaction sponsorship, mitigating errors and improving the overall user experience.

5.1.5 ENS (CometENS)

for Dynamic Service Access & Reduced Memorization: The complexity of managing cryptic blockchain addresses and locating reliable service endpoints adds to cognitive load and memorization difficulties (Section 2.3.6). CometENS leverages the Ethereum Name Service (ENS) to provide human-readable names for both user accounts (via AirAccount) and service nodes registered within SDSS. This enhances discoverability, simplifies configuration for dApps, and reduces the user's need to handle raw addresses, analogous to how web domains simplify internet navigation.

5.2 Countermeasures Against Centralization and Security Risks

SuperPaymaster's decentralized architecture provides inherent advantages over the centralized gas payment solutions analyzed in Section 2.4.1.

5.2.1 Mitigating Monopoly and Cost Inflation

By design, SuperPaymaster fosters a permissionless market where any entity meeting the staking requirements can operate a Paymaster node (Section 3.1.8, 3.6.1). The competitive quoting mechanism (Section 3.5.3) directly counteracts price manipulation and prevents long-term cost inflation associated with market monopolies (Section 2.4.2).

5.2.2 Enhancing Censorship Resistance

Centralized relays can censor transactions based on regulatory pressure (Section 2.4.1). SuperPaymaster's decentralized network of globally distributed nodes, discoverable via SDSS/ENS, significantly increases censorship resistance. Users and dApps have multiple, independent providers to choose from, making it difficult for any single entity or jurisdiction to block transactions universally.

5.2.3 Reducing Transaction Manipulation Risks (MEV)Implications

While MEV remains a fundamental challenge, a decentralized network of competing Paymaster nodes may reduce opportunities for systemic MEV extraction compared to a single dominant centralized relay (Section 2.4.2). Node reputation and transparent operation can further disincentivize malicious behavior. [Optional: Acknowledge that Paymasters themselves don't directly prevent MEV but decentralization changes the landscape].

5.2.4 Improving Security and Privacy

AirAccount integration introduces stronger user authentication via D2FA/biometrics (Section 3.5.4), mitigating risks associated with private key compromise compared to traditional EOA wallets. The SDSS architecture, potentially utilizing TEEs for N2/N3 nodes (Section 3.5.2), aims to minimize privacy leakage risks associated with centralized data aggregation (Section 2.4.3). Data required for sponsorship is minimized, and the decentralized nature prevents single-point-of-failure data breaches.

5.2.5 Trust Model

The SuperPaymaster trust model employs a multi-faceted approach integrating cryptographic verification, economic incentives, reputation, and community governance to ensure security and reliability within a decentralized framework. At its core is a decentralized node mechanism, leveraging on-chain registered independent nodes secured by standard and potentially advanced cryptographic schemes like BLS threshold signatures. A reputation mechanism, potentially adhering to EIP-7562, objectively evaluates node performance based on metrics such as transaction success rates and stake, rewarding reliable service. On-chain smart contracts transparently enforce system rules, verifying node signatures and managing financial flows. A community governance model fosters stakeholder participation in system upgrades and dispute resolution. This interplay creates a positive feedback

loop, termed the "trust flywheel," where high-performing, competitive nodes gain enhanced reputation, attract more users, and solidify their trustworthy position within the ecosystem.

5.3 Implications of Findings

Assuming positive evaluation results of the 'Gas Card' metaphor, this research carries significant implications for both Human-Computer Interaction (HCI) and the Technology Acceptance Model (TAM), alongside Social Learning Theory (SLT) [20]. The validation of this metaphor underscores the efficacy of applying HCI principles, specifically drawing on mental models derived from SLT, to demystify complex blockchain interactions. By leveraging familiar cognitive frameworks, the study demonstrates a tangible pathway to enhance Perceived Ease of Use (PEOU) [24], diminish cognitive load, and consequently foster greater technology acceptance within the Web3 landscape.

Moreover, SuperPaymaster presents a compelling paradigm shift for the blockchain ecosystem, offering a more accessible, competitive, and user-centric gas payment infrastructure. This model not only empowers end-users with simplified gas management and potential cost reductions but also lowers the entry barrier for dApp developers seeking to provide seamless user experiences, thereby fostering innovation and accelerating mainstream blockchain adoption. The OpenPNT/OpenCards initiative further provides communities with novel instruments for engagement and local economic development.

5.4 Limitations

While the SuperPaymaster PoC demonstrates feasibility, several limitations should be acknowledged. The current implementation focuses primarily on EVM-compatible chains; cross-chain interoperability beyond basic ENS naming requires further development. The effectiveness of the Reputation Mechanism relies on robust data collection and potentially complex on-chain logic or trusted off-chain components, which needs more extensive testing under varied network conditions. The economic sustainability of the competitive market and the OpenPNTs model requires deeper analysis and real-world validation to ensure long-term viability and resistance to exploits. Furthermore, the 'Gas Card' metaphor, while beneficial for usability, might imperfectly represent underlying gas price volatility unless Paymaster nodes actively manage this risk, potentially impacting the transparency of actual costs for the user in certain edge cases. The reliance on external components like AirAccount and ENS also introduces dependencies. Finally, this research use a system design, implementation, and evaluation methodology, may not fully capture all potential real-world complexities or attack vectors present in a large-scale deployment.

6 Related Research and Comparison

In recent years, considerable research has investigated the integration of blockchain technology in the account and gas payment and related sector[1,2,3,4,5,15,7,16,17,20,21,22], with relatively fewer studies exploring on HCI, TAM, SLT, and decentralization computing. In this section, we have selected research and works closest to our proposed system. We also investigate the top industry gas payment suppliers with different metrics on their solutions: Pimlico, Alchemy, Stackup, ZeroDev, Coinbase, Biconomy, to show the current state of the industry and compare with the proposed system.

On gasless transaction, we have meta transactions introduced by Ronan Sandford, et al. (2020, July)[1], it needs a relay to sign and pay the gas fee, but it is not a common gas payment solution cause every contract should be modified to support meta trasactions. So Vitalik Buterin introduced ERC4337[2], it has no changes to the existing contract, it just add a new interface to sign the transaction. ERC4337 now is the main standard on Account Abstraction. We have some diffrent solutions on paymaster system, but all of them are inherent from ERC4337 solution.

General Paymaster: Singh, A. K. et al.[3] introduced the basic paymaster mode: verifying paymaster, a mechanism based on ERC4337. It is a paymaster contract that verifies the signature of the off-chain paymaster relay server and then pay

gas for the transaction. This contract "deployed entrypoint contract address and your offchain relay signer address as the constructor parameter values". So it can verify the signature of the paymaster server on-chain. This solution is a part of ERC4337[2],It provides abilities to all contract account to get gas sponsorship or gas payment by another account: paymaster. Bring a great experience to users on gas payment convinient. But it rely on a centralized paymaster server, which may be a single point of failure and risk of censorship, price manipulation and monopolization. And it is only a craft version published by ERC4337 eth-infinitism team[4].

There are two main types of solutions on paymaster systems: centralized company service on ERC4337 and decentralized service on ERC4337. After ERC4337 offical team release the spec, we have a craft version of Gas Sponsorship System. But this solution lack of user interface and service flow, only show the main idea on how to build a gas sponsorship system. Depends on different verify and accounting on other method, we have different paymaster. See the category table:

Pay type	Permission method	Accounting method	Computing method	Oracle method (ERC20)	ERC20 support
Prepay	Use register account with charged balance	Accounting book	Centralized Cloud Server	On-chain Oracle	Pay by ETH in sponsor account
Postpay	Decentralized NFT	Like credit card, Settlement after sponsor	Decentralized Rain Computing	Off-chain Oracle	Pay by ERC20 in your account

Table 4: Paymaster Category

Decentralization: Pimilico[5] is a paymaster production solution based on ERC4337, it provides real interface and service flow to all contract account to get gas sponsorship or gas payment. We get some market data from Bundlebear[6], there are Pimlico, Alchemy, Stackup, ZeroDev, Coinbase, Biconomy and more paymaster services providers on the market. They resolve the basic question: a usable gas payment system based on eth-infinitism team's ERC4337 spec and contract template. honestly, the gas payment market is highly rely on business centerization service.

SuperPaymaster support two basic mode: decentralized NFT verify paymaster, multiple ERC20 support. Anyone can clone and extent the paymaster contract to support more method.

ERC20 Gas Token: Qin Wang [15] also analysed the gas payment mechanism in ERC4337, and gas abstraction: "ERC-20 tokens are accepted as valid payment for these fees, unshackling the restrictions previously confined to ether (ETH)". It is a improvement on gas payment mechanism. ERC4337 solution support ERC20 gas token:"abstracting gas payments allows easy onboarding by 3rd party payments, paying with tokens, cross-chain gas payments", but it does not provide a customized ERC20 to easily pay gas with your community ERC20 tokens in application layer[2]. Most of the companies they just accept their own ERC20 tokens[54]. SuperPaymaster introduce a mechanism named OpenPNTs, make community ERC20 token to be a gas token within several steps.

HCI General: Fröhlich, M.[7] make a deep research and review on HCI in blockchain industry. "With this review, we identify key aspects where interaction design is critical for the adoption of blockchain systems." From trust to understanding motivation, risk, and perception, from wallets to engaging users,using blockchain on use cases to support tools for blockchain, we need so many improvements. On gas payment scenario, there are still two areas need to be improved: User Experience and Service Centralization. As analysis in section 5.3, based on SLT, TAM and HCI[8,9,10,11], we find that we need improve

user experince to reduce the conginition load, reduce the complexity, and make the gas payment process more intuitive and user-friendly. We also find that if we provide a decentralized computing service to paymaster, to improve the trust and lower the risk of censorship, price manipulation and monopolization. Also, to improve the understanding motivation, and PEOU(Perceived Ease of Use)[24], we combine all complex flow and conception into one: gas card with no-feeling gas payment.

Check this table[8,9,10,11]:

	HCI Literature Review Items	HCI Theory Items	Real Scenarios Questions	SuperPaymaster Solution
1	Trust	Evaluation Results	Role of trust	Guarantee by AirAccount: your finger-print and TEE
2	Motivation	Ease of Learning, Efficiency, Memorization, Error Rate, Satisfaction	Social Learning Theory, easy to learn	Just register your ENS name and get a card, your community pay gas for your labor
3	Risk	Cognitive Distribution, Tools, Environment, Social Interaction	Security and Privacy Concerns Manipulation and Censorship Risks Monopoly and Cost Issues	A competitive and decentralized gas sponsor market with affordable quota
4	Perception	User Goals, System State, Execution, Intrinsic Load, Extrinsic Load, Metacognitive Load	Operational Inefficiency Asset Fragmentation Complex On-Chain Transaction Process	ENS name easy to remember and keep gas card (do nothing)
5	Wallets		Hard to use Easy to lost money	Provide SDK for developers
6	Engaging Users	Willingness to Use, Attitude, Perceived Usefulness, Perceived Ease of Use	So many barriers	Retweet and get PNTs for your gas card, negative gas cost with accumulative gas card
7	Specific Application Use Cases	Activity scenarios and pain points	Memorization Difficulties Limited Gas Token Support High Cognitive Load	Enhance 7-1 step, seamlessly gas payment with OpenPNTs
8	Blockchain: Support Tools	Subjects, Tools, Objects, Rules, Communities, Division of Labor	Tool support to be easy	Open-source Contract, SDK, ENS name with community support

Table 5: HCI Analysis

Privacy and Security: Sans, T., & Liu, D. Z. (2024, May)[12] introduce a way to keep the privacy on Account Abstraction on EVM chains. They use Zero Knowledge Proof(ZKP) to ensure the privacy of the data in the environment. On privacy, we have two ways: Trust Execution Enviroment(TEE) and Zero Knowledge Proof(ZKP). TEE is a hardware security environment, it can ensure the privacy computing of the data in the environment[52]. ZKP is a zero knowledge proof, it can ensure the privacy computing of the data in the application[53]. SuperPaymaster use AirAccount, which uses TEE to guarantee the privacy of the data in the transaction flow. But SuperPaymaster can use ZKP add more ensurement if nessesary in the future.

Cost: Lin, Z. research on measurement of Account Abstraction(ERC-4337) on Ethereum[16] tell us: "creating an ERC-4337 account costs 381,489 gas, allowing only 78 accounts per block. Furthermore, a basic ERC-4337 transfer consumes 92,901 gas, which is four times the gas cost of an EOA transfer". So the originall solution is expensive. Fourtunetaly, we have rollup layer2s. Thibault, L. T. [17] tell us, based on ZKP, "fee reduction ranging from 20 times 949 for ETH transfers up to 100 times for ERC20 tokens about 10 times gas reduce". Actually, the gas fee reduce to about 20-30 times in practice[18].

SuperPaymaster use Optimism Layer2 solution with a low gas fee[68,70]. And introduce a competitive gas sponsorship market to get a lower gas fee quote. Further more, we support community gas token(OpenCards/OpenPNTs) to pay gas fee, which you can just retweet and easy to get. Table 6 show us a statistic of the gas fee on different layer2s, cause we pay the gas with Gwei, 10^9 Gwei = 1 ETH and the price of ETH is dynamic, so the gas price on sending ETH or swap tokens actions price below is not a static value, just show a snapshot cost on different blockchains.

Name	Send ETH	Swap Tokens
Metis Network	\$0.04	\$0.18
Loopring	\$0.04	\$0.59
zkSync Era	\$0.07	-
zkSync Lite	\$0.09	\$0.22
Optimism	\$0.09	\$0.18
Arbitrum One	\$0.09	\$0.27
Boba Network	\$0.15	\$0.17
DeGate	\$0.16	\$0.18
StarkNet	\$0.19	\$0.57
Polygon zkEVM	\$0.19	\$2.75
Ethereum	\$1.10	\$5.48

Table 6: Gas Fee Analysis(layer1 and layer2), data source: l2fees.info

Easy-to-use: Even without a theory like HCI[8, 9, 10, 11], we can also draw a conclusion: the key point to the new users to the blockchain: easy-to-use. The new technology should not jump out of the user's daily life. They can get common sense knowledge from others to understand the blockchain operation in Social Learning Theory[20]. Saldivar, J. tell us[19]: "what challenges do first-time blockchain users face when operating blockchain-based applications?", we can know "the most confusing part": the complicated conceptions out of daily life, confuse the new users. "The wording and terms used in the transaction window (e.g., gas fee, gas price, gas limit) were found to be over-complicated by people with limited experience with blockchain technology. " One of the participants verbalized this confusion by asking, "what do you mean by transaction?" (P5). They even know nothing about the gas, transaction, private key, signature and more. SuperPaymaster use a simple gas card metaphor to reduce the abstract the complicated conceptions into one simple concept, which reduce the cognitive load and enhance user experience: gas card. The new users need't know any more new things. Just get a gas card with prepaid balance, and use it to pay gas fee. We use NFT(Non Fungible Token)/SBT(Soul Bound Token) to be the gas card credential, the contract account be the community PNTs account, the ERC20 token be the gas token and all into a seamless flow.

DApps: Glomann, L.[21] get four issues to the DApps to mass adoption are: The motivation to change, the onboarding challenge, the usability problem and the feature problem. Obviously, the gas payment process is a necessary and initial barrier to all the DApps users, and if we provide a unique and decentralized, developer friendly interface(like paymaster.aastar.eth) and SDK, it can reduce the onboarding challenge and improve the usability problem. Krug, S[22]. tell use, as an application, we need to focus on learnability, convenience and "Don't Make Me Think". SuperPaymaster practice the rules: Self-evident: no feeling gas payment with gas card concept; Obvious: use ENS to simplify the wallet address; Self-explanatory: use NFT to represent the gas card.

Table 7: Multi-dimensional comparison analysis table:

Field	Ronan S et [1]	Vitalik et [2] [4]	Singh, et.[3]	Qin Wang [15]	Lin, Z. et. [16]	Thibault, L [17]	Pimlico [5]	Alchemy [18]
Type	Industry	Industry	Accademic	Accademic	Accademic	Accademic	Industry	Industry
Purpose	EIP2771 to support meta transaction	ERC4337 to create account abstraction non-modify and surplus on current tech stack	Implementate ERC4337 solution	Discuss gas token on ERC4337	Discuss gas cost on Layer1, Layer2	Research on Layer2 rollup	Full ERC4337 implementation on paymaster(bundler)	Combine in one solution account(with gas sponsor bundler)
Solution	Need every contract customize modification	alt-mempool framework with basic account abstraction flow on entrypoint, bundler and paymaster	verifying paymaster on relay signature and on-chain verify	easy onboarding by 3rd party payments, paying with tokens	Use Layer2 to reduce the gas cost	Try to reduce gas cost, some Layer2 support customize ERC20 gas token	Centralized API key with verifying and ERC20 paymaster service	A top busir gas sponsor service
Solution Account	EOA	Contract account demo	Contract account	Contract account	Contract account	EOA	Contract account	Contract account
Solution Relay	✗	✗	✓	✓	✓	✗	✓	✓
Solution Simple	✗	✗	✗	✗	✗	✗	✗	✗
Solution Time/Efficiency	✗	✗	✗	✗	✗	✗	✗	✗
Solution Customize ERC20	✗	✗	✗	✗	✗	✓	✓	✓
Cost Direct Cost	Low	High	High	High	Medium	Medium	Medium	Medium
Usability&UX:HCI HCI:Cognitive Load	High	High	High	High	High	High	Medium	Medium
Usability&UX No Memorization	✗	✗	✗	✗	✗	✗	✗	✓
Usability&UX HCI:Efficiency	✗	✗	✗	✗	✗	✗	✗	✓
Usability&UX HCI:Fault Tolerance	✗	✗	✗	✗	✗	✗	⚠	⚠

Field	Ronan S et [1]	Vitalik et [2] [4]	Singh, et.[3]	Qin Wang [15]	Lin, Z. et. [16]	Thibault, L [17]	Pimlico [5]	Alchemy [18]
Decentralization TAM:E&E Gulf	✗	✗	✗	✗	✗	✗	✗	✓
Decentralization SLT: Society Learning	✗	✗	✗	✗	✗	✗	✗	⚠
Decentralization Support	✗	✗	✗	✗	✗	✗	✗	✗
Comment	EIP2771	AA	AA	Layer2	Layer2	Rollup/Layer2	First business provider	Total solution provider

Table 8: Multi-dimensional comparison on industry general area table

Feature/Solution	Pimlico	ZeroDev	Alchemy	Biconomy	Coinbase	Particle Network	Stack
Main Features	Bundler and Paymaster Infrastructure	Modular Smart Accounts and Plugin System	Full-stack AA Toolkit	Modular Cross-chain Smart Accounts	Ecosystem-specific AA Solution	Cross-chain Unified Account and Balance	Enterprise grade Account Solution
Core Products	Alto Bundler, Verifying/ERC20 Paymaster	Kernel Smart Account, Plugin System	Account Kit, Rundler, Gas Manager	Modular Smart Account, MEE	Verifying Paymaster, Bundler API	Universal Accounts, Omnichain Paymaster	Enterprise Smart Paymaster API
Smart Account Standard	Universal	ERC-7579	ERC-6900	ERC-7579	Universal	Proprietary+ERC-4337	Universal
Cross-chain Capability	Medium (Multi-chain Deployment)	Medium (Multi-chain Deployment)	Medium (Multi-chain Deployment)	High (MEE)	Low (Base-focused)	Very High (Universal Account)	Medium (Multi-chain Deployment)
ERC20 Gas Payment	Full Support	Full Support	Full Support	Full Support	Partial Support	Full Support	Full Support
Gas Sponsorship Method	API Key, Webhook Policies	Meta-infrastructure Proxy	Gas Manager, Policy Engine	Paymaster API, Policies	Base Ecosystem Optimized	Chain Abstraction Layer Sponsorship	API Key Enterprise Policy
Open Source Status	Highly Open Source	Highly Open Source	Partially Open Source	Highly Open Source	Partially Open Source	Progressively Opening	Partially Open Source
Development Complexity	Low-Medium	Medium	Medium-High	Medium	Low	Medium	Medium-High
Core SDK	permissionless.js	@zerodev/sdk	@alchemy/aa-core	@biconomy/account	@coinbase/various-sdks	@particle-network/aa-sdk	usercentric
PassKey Support	Via Integration	Native Support	Via Plugins	Via Modules	Via Integration	Via Auth Service	Native Support
Suitable Use Cases	Infrastructure Support	Consumer Apps, Gaming	Enterprise Apps, Complex DeFi	Cross-chain Apps, DeFi	Base Ecosystem Apps	Cross-chain Apps, Bitcoin L2	Enterprise Collaboration Apps

User Count (BundleBear)	N/A (Infrastructure)	~900K Accounts	~7.3M Light Accounts	~224K Accounts	~36K Accounts	~200K+ on Bitcoin L2	~34K Acco
Advantages	High Performance, Reliability	Modularity, Gas Efficiency	Complete Toolkit, Enterprise Support	Cross-chain Capability, Modularity	Base Ecosystem Integration, Ease of Use	Cross-chain Unified Account, Bitcoin Support	Enter Secu Colla Featu
Disadvantages	Infrastructure Only	Plugin System Learning Curve	Some Features Not Open Source	Infrastructure Dependencies	Ecosystem Limitations	Newer Solution	Enter Focu Gene

7 Conclusion

Widespread blockchain adoption remains hindered by significant user challenges, primarily high gas fees, complex payment processes, and poor user experiences stemming from overly technical and centralized solutions, which deter new user onboarding. Addressing these core usability and centralization obstacles, alongside related issues in account management and security, is crucial for unlocking the full potential of decentralized applications and services.

SuperPaymaster is a novel system designed for a seamless ("gasless" perception), user-friendly, developer-accessible, decentralized, secure, and scalable gas payment system. A key contribution is the proposal of a fully competitive gas sponsorship market built upon this system. We introduce the OpenPNTs and OpenCards system, leveraging the familiar "Gas Card" metaphor. This maps the complex gas payment process onto simpler "topping up" and "paying" concepts, significantly reducing cognitive load and enhancing user experience. Furthermore, AirAccount integration provides robust decentralized second-factor authentication (D2FA), boosting security and usability during gas payment interactions. To address cryptic wallet and API addresses, CometENS enhances their readability and memorability. Underpinning the architecture is the Standardized Decentralized Service System (SDSS), facilitating relay server operation on decentralized compute nodes. This mitigates risks of price manipulation and censorship in centralized gas sponsorship services, aligning with DePIN principles through open-source solutions and permissionless registration contracts.

For future work, we plan to explore the implementation of cross-chain SuperPaymaster capabilities, including investigating interoperability with emerging Account Abstraction standards like RIP-7560 and integration with EIP-7702 to potentially support gas sponsorship for traditional Externally Owned Accounts (EOAs)[55, 69]. Further research directions include enhancing the relay server framework by integrating AI and security policy vector models to provide more sophisticated and proactive transaction security assurances. Additionally, we intend to conduct deeper analyses of community economic models enabled by OpenPNTs/OpenCards, perform large-scale deployment and usability studies of the SDSS architecture, and continue refining the system to better serve blockchain public goods and contribute positively to the evolving human digital future.

8. Acknowledgments

This research was financed by the Plancker^ Community, and development was supported by the AAStar Team which was a subsidiary of Plancker^.

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Appendices

Includes: supplementary information table, etc.

Core Components

While the infrastructure layer, comprising AirAccount and SDSS and more, ensures sovereign identity and secure data storage for all ecosystem participants, the COS72 framework provides developers with the specific architectural blueprint needed to rapidly deploy new DApps that leverage the HyperCapital asset.

Layer	Describe	Repo
Infrastructure	SuperPaymaster: a universal open source paymaster contract for account abstraction.	https://github.com/AAStarCommunity/SuperPaymaster
Infrastructure	SDS: an architecture for decentralized service sponsor system.	https://github.com/AAStarCommunity/SDSS
Infrastructure	AirAccount: an open source account abstraction contract.	https://github.com/AAStarCommunity/AirAccount
Infrastructure	AirAccount-Rust-Relay: an open source relay server for AirAccount.	https://github.com/AAStarCommunity/AirAccount-Rust-Relay
Infrastructure	OpenPNTs: an open source token solution for community sustainability	https://github.com/AAStarCommunity/OpenPNTs
Infrastructure	CometENS: an open source ENS for AirAccount and SDSS.	https://github.com/AAStarCommunity/CometENS
framework	HexagonWarrior: Multi OS client framework	https://github.com/AAStarCommunity/HexagonWarrior-Tauri
framework	COS72 SDK and Demo: A quick Nodejs SDK and a demo to show features	https://github.com/AAStarCommunity/AAStar_SDK

AAStar Team

- It is a team focusing on Ethereum ecosystem, core members talked AA early with Vitalik in 2022 Moutainegro, and now working on AA and more infra over 3+ years.

Plancker^ Community

- It is a community to help Ethereum builders, initiated by Nicolas and more guys, they help many Open-source projects in Ethereum. Incubated AAStar from 2022 to 2024 Nov.